

Restricting externalization

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Abstract

It is standardly assumed that linear order is asymmetric and that Vocabulary Insertion is one-to-one. We argue that both assumptions should be abandoned: linear order is antisymmetric, thereby allowing reflexivity, and Vocabulary Insertion is many-to-many. We show that these assumptions allow for straightforward accounts of (i) the linearization of simultaneity, (ii) the generation of multiple exponence, and (iii) the insertion of portman-teaux, all without adopting additional morphological primitives. Ultimately, we advance a research program called Intermodular Morphology, which seeks to account for morphological phenomena using only the syntax, phonology, and intermodular (PF) operations that translate between them.

Keywords: linearization, antisymmetry, simultaneity, Vocabulary Insertion, portman-teaux, multiple exponence, Intermodular Morphology

1 Introduction

In this article, we argue for changing two common assumptions about the externalization of syntactic structure. First, we argue that linear order is antisymmetric rather than asymmetric, contra Kayne (1994) and subsequent authors.¹ Second, we argue that Vocabulary Insertion

is not one-to-one but many-to-many, with the tendency toward one-to-one mappings driven by external factors. By changing our assumptions in this way, we eliminate the need for several morphological operations, simplifying the process of externalization.

The article proceeds as follows. Section 2 presents the theoretical motivation for these proposals. Section 3 then demonstrates the empirical advantages of these proposals, showing that a many-to-many theory of Vocabulary Insertion allows for multiple exponence, that an antisymmetric theory of linear order straightforwardly derives simultaneity phenomena (e.g., grammatical tone), and that together, they allow for the insertion of portmanteaux without additional morphology-specific operations. Section 4 concludes.

2 Theoretical motivation

2.1 Motivating antisymmetry

Following Kayne (1994), it is common to model word order as a binary relation, that is, a set of ordered pairs. Kayne (1994) argues that this relation—henceforth referred to as the *precedence relation*—must be a (*strict*) *linear order* to be well-formed. In other words, it must be transitive (1), total (2) and asymmetric (3). Within the field of linguistics, this latter property is often referred to as *antisymmetry*, particularly when discussing linear order (e.g., Kayne, 1994). However, as the terms are typically used outside of linguistics, antisymmetry (4) differs from asymmetry in that only asymmetry entails irreflexivity (5).

- (1) Let R be a binary relation over a set S . Then R is *transitive* iff for all $x, y, z \in S$, if $\langle x, y \rangle \in R$ and $\langle y, z \rangle \in R$ then $\langle x, z \rangle \in R$. Informally, if R relates x to y , and R relates y to z , then R also relates x to z .
- (2) Let R be a binary relation over a set S . Then R is *total* (or *connected*) iff for all $x, y \in S$ such that $x \neq y$, either $\langle x, y \rangle \in R$ or $\langle y, x \rangle \in R$. Informally, if x and y are distinct, then

R relates x to y or R relates y to x .

- (3) Let R be a binary relation over a set S . Then R is *asymmetric* iff for all $x, y \in S$, if $\langle x, y \rangle \in R$ then $\langle y, x \rangle \notin R$. Informally, if R relates x to y , then R does not relate y to x .
- (4) Let R be a binary relation over a set S . Then R is *antisymmetric* iff for all $x, y \in S$, if $\langle x, y \rangle \in R$ and $\langle y, x \rangle \in R$, then $x = y$. Informally, if R relates x to y and y to x , then x and y are the same entity.
- (5) Let R be a binary relation over a set S . Then R is *irreflexive* iff for all $x \in S$, $\langle x, x \rangle \notin R$. Informally, R does not relate x to itself.

We argue that the precedence relation is better modeled as antisymmetric than asymmetric. This proposal is motivated by changes to the theoretical landscape since the publication of Kayne 1994—in particular, the reframing of movement as (re-)Merge. Whenever a constituent moves, it comes to c-command its base position and therefore itself (reflexive c-command), and both c-commands and is c-commanded by any constituent it crosses over (symmetric c-command; cf. Johnson, 2020 for a proposal where this does not hold). Under many approaches to the syntax-phonology mapping, this leads to corresponding symmetry and reflexivity during linearization (see Nunes, 2004 for discussion).

However, reflexivity is trivial to resolve. Under a copy deletion approach (e.g., Nunes, 2004), we simply delete all but one copy of a constituent. For the purposes of avoiding reflexivity—preventing a constituent from preceding itself—it does not matter which copies we delete. Likewise, under an approach where we delete ordered pairs from the precedence relation (Malanoski, 2024; see also Johnson, 2012), we simply delete the reflexive ordered pair. In general, resolving reflexivity has no bearing on the rest of the linearization process. Given that reflexivity is ubiquitous in syntax, arising whenever displacement occurs, and trivial to resolve, why bother resolving it? In other words, why not let the precedence relation be antisymmetric rather than

asymmetric, thus allowing reflexivity? In section 3.3 and section 3.4, we demonstrate that this proposal can account for simultaneity phenomena that challenge the asymmetry-based approach.

2.2 Motivating many-to-many Vocabulary Insertion

Contemporary theories of morphology standardly assume that Vocabulary Insertion—the mapping of syntactic units onto their phonological exponents—is one-to-one, so that each syntactic unit has exactly one exponent and each exponent realizes exactly one syntactic unit (although theories disagree on which syntactic unit is the target of Vocabulary Insertion).² These approaches to Vocabulary Insertion directly encode the *principle of biuniqueness*, “an idealized approach to the structure of words in which one form corresponds to one meaning” (Harris, 2016, 3). However, this principle is routinely violated in natural language (Harris, 2016). For example, many languages contain portmanteaux, single exponents that encode multiple grammatical categories. Conversely, many languages demonstrate multiple exponence, in which multiple exponents encode the same grammatical category. An example is given in (6), which shows multiple exponence of the class marker *y-* in a Batsbi (Nakh-Daghestanian: Georgia) sentence (Harris, 2016, 2; irrelevant details removed from gloss). These phenomena exemplify a many-to-one and one-to-many mapping onto exponents, respectively.

- (6) *y-ox-y-o-y-an^w* *kʰab*
 CM-rip-CM-PRES-CM-EVID dress.ABS
 ‘Evidently she is ripping the dress.’

Because they assume the principle of biuniqueness, morphological theories often must propose additional theoretical tools to accommodate such phenomena. Distributed Morphology (Halle & Marantz, 1993), which assumes a one-to-one mapping between terminal nodes and their exponents, addresses these issues by proposing morphological operations that alter the phrase structure. These include Fusion, which combines two terminal nodes into one, thereby

allowing for the insertion of portmanteaux; and Fission, which allows multiple exponents to be inserted for a single terminal node, thereby allowing for certain types of multiple exponence.³ However, such operations are conceptually undesirable. In general, the postsyntactic operations of Distributed Morphology are theoretical primitives, constituting an arguably undesirable complication of the theory of Language. Moreover, treating these operations as primitive raises the question of why they exist in Language and how they arose. Finally, although these operations manipulate syntactic structures, they are not constrained by syntactic principles such as the No-Tampering Condition (Chomsky, 2008), creating an unconstrained “second syntax” (Collins & Kayne, 2023).

Nanosyntax (Caha, 2009; Starke, 2009) eschews most of the postsyntactic operations of Distributed Morphology. Moreover, researchers working within Nanosyntax typically adopt either non-terminal spell-out, where Vocabulary Insertion targets phrasal nodes, or spanning, where Vocabulary Insertion targets sequences of heads (Taraldsen, 2018). In both cases, the unit of insertion may be larger than a single terminal node, and so portmanteaux can be inserted without adopting an operation such as Fusion that combines multiple terminal nodes into one. Nevertheless, Nanosyntax faces challenges of its own. In general, those who assume non-terminal spell-out typically also propose “spell-out driven movement” to derive the structural configurations for Vocabulary Insertion (Baunaz & Lander, 2018). Unlike other syntactic movement, spell-out driven movement is not driven by formal features. Moreover, it is typically assumed not to leave traces. This means that it alters the existing syntactic structure, in violation of the No-Tampering Condition. Thus, it cannot be considered syntactic movement, given that it does not obey syntactic principles. Nevertheless, it takes syntactic representations as input and produces syntactic representations as output. In this sense, spell-out driven movement constitutes a second syntax in Collins & Kayne’s (2023) sense, a way of manipulating syntactic structures without obeying syntactic principles. As for spanning, it introduces a new theoretical primitive: outside of its role in Vocabulary Insertion, “the notion of span has no

status in current theory” (Svenonius, 2016, 204). In short, Nanosyntax’s treatment of portmanteaux faces conceptual hurdles of its own. Moreover, it is not obvious how to account for multiple exponence under Nanosyntax. One solution is to treat multiple exponence as involving different functional heads. For example, Vanden Wyngaerd et al. (2021) argue that in *better*, where comparative meaning is realized both by the suppletive form *bett-* and by the comparative suffix *-er*, *bett-* realizes one comparative projection (as well as the *aP*) while *-er* realizes a separate comparative projection. However, it is not obvious that this approach generalizes to all types of multiple exponence. Cases such as (6) seem especially problematic: unless we propose that there are multiple non-adjacent class marker projections—an unlikely state of affairs—it is unclear how to derive the three non-adjacent class markers. Furthermore, the multiple-projection approach would leave unexplained why each projection is realized by the same exponent.

We argue that the conceptual issues faced by Distributed Morphology and Nanosyntax arise in part from their commitment to the principle of biuniqueness. We will show below that by abandoning this principle as a definitional trait of Vocabulary Insertion, we can generate multiple exponence and portmanteaux without a “second syntax” or a proliferation of theoretical primitives. In place of the principle of biuniqueness, we propose that the mapping from syntactic units to phonological units is many-to-many. That is, the mapping characterizing Vocabulary Insertion is both one-to-many and many-to-one.

3 Empirical consequences

In this section, we present empirical arguments in support of our proposals. We begin by presenting our theoretical proposal in section 3.1. We then show in section 3.2 that by allowing Vocabulary Insertion to be many-to-many, we can generate multiple exponence (a one-to-many mapping) without assuming operations such as Fission or Fusion. In section 3.3, we show that by adopting an antisymmetric conception of linear order, we can allow straightforwardly for

the linearization of simultaneous material. In section 3.4, we show that these proposals together allow for the insertion of portmanteaux and suppletion (many-to-one mappings). Finally, section 3.5 demonstrates that our proposal can indeed generate many-to-many mappings.

3.1 Theoretical proposal

We assume that linearization is post-syntactic (Chomsky, 1995). We also adopt late insertion of lexical items (Halle & Marantz, 1993), and assume that this follows linearization (Arregi & Nevins, 2012; Ostrove, 2018). We refer to this process as Vocabulary Insertion, as above, and use the term *Vocabulary Item* interchangeably with *exponent*. Since linearization precedes Vocabulary Insertion, it generates a binary relation over syntactic nodes (terminal nodes, by assumption). We call this the u(nderlying)-precedence relation P_u (7). The u-precedence relation is then mapped onto the s(urface)-precedence relation P_s , a binary relation over Vocabulary Items (8). We remain agnostic on the algorithm that generates the u-precedence relation. In examples taken from other sources, we adopt the algorithm assumed by the author(s), unless otherwise noted.

(7) A *u-precedence relation* is a binary relation over terminal nodes.

(8) An *s-precedence relation* is binary relation over Vocabulary Items.

Because the requirements that a precedence relation be transitive, total and antisymmetric are output conditions, they need only hold of the s-precedence relation, rather than the phonologically contentless u-precedence relation.

It is Vocabulary Insertion that replaces the u-precedence relation with the s-precedence relation. Specifically, for all $\langle X, Y \rangle \in P_u$, Vocabulary Insertion adds $\langle x, y \rangle$ to P_s , where x is the Vocabulary Item selected to realize node X and y is the Vocabulary Item selected to realize node Y (this will be amended slightly in section 3.2). It is possible that $x = y$, for example if a portmanteau is available that realizes both X and Y (see section 3.4). We say that P_s realizes

P_u and that $\langle x, y \rangle$ realizes $\langle X, Y \rangle$. Throughout this article, we use uppercase variable names for terminal nodes and lowercase variable names for Vocabulary Items.

The relationship between the u-precedence and s-precedence relations is constrained by the *Principle of Maximality* (9), named after Optimality Theory’s MAX (Prince & Smolensky, 2004). This constraint essentially forbids deletions of ordered pairs from the s-precedence relation.

- (9) Principle of Maximality (strong version): Let P_u be a u-precedence relation that is mapped onto an s-precedence relation P_s . Then for all $\langle X, Y \rangle \in P_u$ there exists $\langle x, y \rangle \in P_s$ such that x is an exponent of X and y is an exponent of Y .

(9) is compatible with approaches to linearization where re-merge does not cause ordering contradictions (e.g., de Vries, 2009; Sheehan, 2013). However, it poses a problem for approaches where linearization sometimes requires repair (e.g., Nunes, 2004; Malanoski, 2024). Because the requirements of transitivity, totality and antisymmetry only apply to the s-precedence relation, it cannot be known until the s-precedence relation is generated whether repair is necessary. To repair antisymmetry violations, we must delete ordered pairs. In other words, if we have both $\langle x, y \rangle$ and $\langle y, x \rangle$ where $x \neq y$, the only way to generate an antisymmetric relation is to delete one of the ordered pairs. However, as formulated in (9), the Principle of Maximality militates against the deletion of ordered pairs from the s-precedence relation. Furthermore, we cannot resolve this by deleting ordered pairs from the u-precedence relation without look-ahead: antisymmetry is a requirement on the s-precedence relation, not the u-precedence relation, so it is unknown before we generate the s-precedence relation whether repair will be necessary.⁴

One possible solution is to treat (9) as a violable constraint, allowing the minimal number of repairs to resolve an ordering conflict. A second solution is to redefine the Principle of Maximality along the lines of (10) so that it only allows necessary deletions.

- (10) **Principle of Maximality (weak version):** Let P_u be a u-precedence relation that is mapped onto an s-precedence relation P_s . Then for all $\langle X, Y \rangle \in P_u$ such that $\langle Y, X \rangle \notin P_u$, there exists $\langle x, y \rangle \in P_s$ such that x is an exponent of X and y is an exponent of Y .

To illustrate, suppose we have a u-precedence relation P_u with $\langle X, Y \rangle, \langle Y, X \rangle \in P_u$, and that Vocabulary Insertion inserts some Vocabulary Item x for X and some Vocabulary Item y for Y , where $x \neq y$. This gives an s-precedence relation $P_s = \{\langle x, y \rangle, \langle y, x \rangle\}$. This s-precedence relation is not antisymmetric. However, we can repair this under the weak Principle of Maximality. For example, we can repair P_s by deleting $\langle x, y \rangle$: $\langle x, y \rangle$ realizes $\langle X, Y \rangle$, but $\langle Y, X \rangle \in P_u$, so the weak Principle of Maximality does not protect $\langle x, y \rangle$. The same reasoning shows that $\langle y, x \rangle$ is also deletable.⁵ In general, the weak Principle of Maximality only permits deletions that resolve antisymmetry violations. For expositional purposes, we will assume the inviolable weak Principle of Maximality rather than the violable strong Principle of Maximality in order to avoid having to compare different linearizations. We leave comparison of these and other potential approaches to future work.⁶

In addition to the Principle of Maximality, which governs the relationship between the u-precedence and s-precedence relations, we propose a Precedence Mapping Principle (11), which governs the relationship between the s-precedence relation and the output string. Note that something like the Precedence Mapping Principle is a conceptual necessity: the s-precedence relation needs to be mapped onto precedence in order to produce a string.

- (11) *Precedence Mapping Principle:* Let P_s be an s-precedence relation. For all Vocabulary Items x, y , if $\langle x, y \rangle \in P_s$, then x precedes y .

Because we allow reflexivity in the s-precedence relation (recall that this is the consequence of adopting antisymmetry rather than asymmetry; see section 2.1), precedence must be defined in a way that allows reflexivity, as in (12). Note that we intend an inclusive interpretation of *or*; this will be relevant to the discussion of autosegmental phenomena in section 3.3. On analogy

with the distinction between subsethood and proper subsethood, we refer to the more traditional notion of precedence as proper precedence (13).

(12) Given lexical items x, y , x *precedes* y ($x \leq y$) if x is pronounced before or at the same time as y .

(13) Given lexical items x, y , x *properly precedes* y ($x < y$) if x is pronounced before y .

It is crucial that the Precedence Mapping Principle is a material implication, not a biconditional, so that it is not the case that if an occurrence of x precedes an occurrence of y , then $\langle x, y \rangle \in P_s$. This is necessary because any Vocabulary Item (improperly) precedes itself: a word is pronounced at the same time as itself.⁷ To illustrate, suppose we have an s-precedence relation $P_s = \{\langle x, y \rangle\}$. By the Precedence Mapping Principle, this maps onto the string xy . However, in this string, x (improperly) precedes itself, but $\langle x, x \rangle \notin P_s$, so the converse of the Precedence Mapping Principle does not hold for this example. To the extent that such cases are typical, this indicates that the Precedence Mapping Principle must be a material implication rather than a biconditional: the converse does not hold.⁸

3.2 Multiple exponence

By allowing Vocabulary Insertion to be potentially many-to-many, the present proposal can also account for multiple exponence, “the occurrence of multiple realizations of a single feature, bundle of features, or derivational category in more than one position in a domain” (Caballero & Harris, 2012, 165). We take multiple exponence to include the phenomenon sometimes known as “multiple copy spell-out” (e.g., Kandybowicz, 2008), where the same word or phrase appears multiple times in an utterance. (14) gives an example from dialectal German in which the *wh*-word *wen* appears twice.

(14) *wen glaubst du wen Hans sieht?*
 who thinks you who Hans sees

‘Whom you think Hans sees?’ (McDaniel, 1986, 183)

We view “multiple copy spell-out” as multiple exponence whose domain is (potentially) larger than the word. If we reject the separation between morphology and syntax (e.g., Baunaz & Lander, 2018; Collins & Kayne, 2023), then unifying repetition of exponents above and below the word level in this way is desirable, if not necessary.

If Vocabulary Insertion is many-to-many, then Vocabulary Insertion is not forced to insert one exponent for each head. This requires a slight modification of our description of Vocabulary Insertion. In section 3.1, we said that given a u-precedence relation P_u , Vocabulary Insertion adds $\langle x, y \rangle$ to the s-precedence relation P_s for all $\langle X, Y \rangle \in P_u$, where x is the Vocabulary Item selected to realize X and y is the Vocabulary Item selected to realize Y . If Vocabulary Insertion can insert multiple exponents, then it is incorrect to say that x and y are *the* Vocabulary Items inserted for X and Y . Rather, a more accurate description of Vocabulary Insertion is the following: for all $\langle X, Y \rangle \in P_u$, Vocabulary Insertion adds $\langle x, y \rangle$ to P_s for all x inserted for X and for all y inserted for Y .⁹

Given that we allow Vocabulary Insertion to be many-to-many, we view the preference for single exponence as a tendency caused by other factors.¹⁰ First, a principle of Pronunciation Economy may directly favor the insertion of the fewest exponents possible (Fanselow & Ćavar, 2001). After all, it is easier for the speaker to say less. There may also be a parsing benefit to single exponence. In a sentence with only single exponence, every Vocabulary Item realizes different syntactic nodes. But when multiple exponence is introduced, the problem arises of determining whether two Vocabulary Items represent the same or different syntactic nodes. By muddling the relationship between syntactic nodes and Vocabulary Items, multiple exponence makes it more difficult to assign syntactic structure to a string. Moreover, if deletion of ordered pairs from the s-precedence relation is subject to economy, then certain types of multiple exponence will be disfavored for requiring more ordered pairs to be deleted. In particular, fully

superfluous multiple exponence (Caballero & Harris, 2012), where multiple exponents realize the exact same features, typically requires more ordered pairs to be deleted than single exponence. Let us illustrate, taking the opportunity to explain how multiple exponence arises under the present proposal.

To begin, consider the Standard German counterpart of (14), presented in (15).

- (15) wen glaubst du dass Hans sieht?
who thinks you that Hans sees
‘Whom do you think Hans sees?’

We assume for (14) and (15) the structure in Figure 1.¹¹ For clarity, we include exponents as well as nodes in this and subsequent trees. However, it should be kept in mind that having adopted late insertion, exponents are not taken to be present in the syntax. Additionally, for expositional purposes we ignore vP and the internal structure of the DPs in Figure 1.

Assuming for expositional purposes that asymmetric c-commands maps onto precedence (Kayne, 1994), linearizing the structure in Figure 1 produces the u-precedence relation in Table 1. In producing (15), we assume that Vocabulary Insertion inserts exponents according to the mappings in (16), which will map the u-precedence relation onto the s-precedence relation in Table 2.

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$\langle \text{DP}_1, \text{V}_1 \rangle$	$\langle \text{DP}_1, \text{T}_1 \rangle$	$\langle \text{DP}_1, \text{C}_1 \rangle$	$\langle \text{DP}_1, \text{DP}_2 \rangle$	$\langle \text{DP}_1, \text{DP}_1 \rangle$	$\langle \text{DP}_1, \text{C}_2 \rangle$	$\langle \text{DP}_1, \text{DP}_3 \rangle$	$\langle \text{DP}_1, \text{V}_2 \rangle$	$\langle \text{DP}_1, \text{T}_2 \rangle$
$\langle \text{V}_1, \text{V}_1 \rangle$	$\langle \text{V}_1, \text{T}_1 \rangle$	$\langle \text{V}_1, \text{C}_1 \rangle$	$\langle \text{V}_1, \text{DP}_2 \rangle$	$\langle \text{V}_1, \text{DP}_1 \rangle$	$\langle \text{V}_1, \text{C}_2 \rangle$	$\langle \text{V}_1, \text{DP}_3 \rangle$	$\langle \text{V}_1, \text{V}_2 \rangle$	$\langle \text{V}_1, \text{T}_2 \rangle$
$\langle \text{T}_1, \text{V}_1 \rangle$	$\langle \text{T}_1, \text{T}_1 \rangle$	$\langle \text{T}_1, \text{C}_1 \rangle$	$\langle \text{T}_1, \text{DP}_2 \rangle$	$\langle \text{T}_1, \text{DP}_1 \rangle$	$\langle \text{T}_1, \text{C}_2 \rangle$	$\langle \text{T}_1, \text{DP}_3 \rangle$	$\langle \text{T}_1, \text{V}_2 \rangle$	$\langle \text{T}_1, \text{T}_2 \rangle$
$\langle \text{C}_1, \text{V}_1 \rangle$	$\langle \text{C}_1, \text{T}_1 \rangle$		$\langle \text{C}_1, \text{DP}_2 \rangle$	$\langle \text{C}_1, \text{DP}_1 \rangle$	$\langle \text{C}_1, \text{C}_2 \rangle$	$\langle \text{C}_1, \text{DP}_3 \rangle$	$\langle \text{C}_1, \text{V}_2 \rangle$	$\langle \text{C}_1, \text{T}_2 \rangle$
$\langle \text{DP}_2, \text{V}_1 \rangle$	$\langle \text{DP}_2, \text{T}_1 \rangle$			$\langle \text{DP}_2, \text{DP}_1 \rangle$	$\langle \text{DP}_2, \text{C}_2 \rangle$	$\langle \text{DP}_2, \text{DP}_3 \rangle$	$\langle \text{DP}_2, \text{V}_2 \rangle$	$\langle \text{DP}_2, \text{T}_2 \rangle$
				$\langle \text{C}_2, \text{DP}_1 \rangle$		$\langle \text{C}_2, \text{DP}_3 \rangle$	$\langle \text{C}_2, \text{V}_2 \rangle$	$\langle \text{C}_2, \text{T}_2 \rangle$
				$\langle \text{DP}_3, \text{DP}_1 \rangle$			$\langle \text{DP}_3, \text{V}_2 \rangle$	$\langle \text{DP}_3, \text{T}_2 \rangle$
				$\langle \text{V}_2, \text{DP}_1 \rangle$			$\langle \text{V}_2, \text{V}_2 \rangle$	$\langle \text{V}_2, \text{T}_2 \rangle$
				$\langle \text{T}_2, \text{DP}_1 \rangle$			$\langle \text{T}_2, \text{V}_2 \rangle$	

Table 1: The u-precedence relation for the structure in Figure 1.

(16) Vocabulary mapping for (15).

- a. $DP_1 \rightarrow \textit{wen}$
- b. $V_1 \rightarrow \textit{glaub-}$
- c. $T_1 \rightarrow \textit{-t}$
- d. $C_1 \rightarrow \emptyset$
- e. $DP_2 \rightarrow \textit{du}$
- f. $C_2 \rightarrow \textit{dass}$
- g. $DP_3 \rightarrow \textit{Hans}$
- h. $V_2 \rightarrow \textit{sieb-}$
- i. $T_2 \rightarrow \textit{-t}$

This s-precedence relation is not antisymmetric—for example, it contains both $\langle \textit{wen}, \textit{du} \rangle$ and $\langle \textit{du}, \textit{wen} \rangle$ —and thus requires repair. To generate the surface form (15), the underlined ordered pairs are deleted, producing the relation in Table 3. This s-precedence relation will generate the surface form in (15).

$\langle \text{wen}, \text{glaub-} \rangle$	$\langle \text{wen}, -st \rangle$	$\langle \text{wen}, C_1 \rangle$	$\langle \text{wen}, du \rangle$	$\langle \text{wen}, \text{wen} \rangle$	$\langle \text{wen}, \text{dass} \rangle$	$\langle \text{wen}, \text{Hans} \rangle$	$\langle \text{wen}, \text{sieh-} \rangle$	$\langle \text{wen}, -t \rangle$
$\langle \text{glaub-}, \text{glaub-} \rangle$	$\langle \text{glaub-}, -st \rangle$	$\langle \text{glaub-}, C_1 \rangle$	$\langle \text{glaub-}, du \rangle$	$\langle \text{glaub-}, \text{wen} \rangle$	$\langle \text{glaub-}, \text{dass} \rangle$	$\langle \text{glaub-}, \text{Hans} \rangle$	$\langle \text{glaub-}, \text{sieh-} \rangle$	$\langle \text{glaub-}, -t \rangle$
$\langle -st, \text{glaub-} \rangle$	$\langle -st, -st \rangle$	$\langle -st, C_1 \rangle$	$\langle -st, du \rangle$	$\langle -st, \text{wen} \rangle$	$\langle -st, \text{dass} \rangle$	$\langle -st, \text{Hans} \rangle$	$\langle -st, \text{sieh-} \rangle$	$\langle -st, -t \rangle$
$\langle C_1, \text{glaub-} \rangle$	$\langle C_1, -st \rangle$		$\langle C_1, du \rangle$	$\langle C_1, \text{wen} \rangle$	$\langle C_1, \text{dass} \rangle$	$\langle C_1, \text{Hans} \rangle$	$\langle C_1, \text{sieh-} \rangle$	$\langle C_1, -t \rangle$
$\langle du, \text{glaub-} \rangle$	$\langle du, -st \rangle$			$\langle du, \text{wen} \rangle$	$\langle du, \text{dass} \rangle$	$\langle du, \text{Hans} \rangle$	$\langle du, \text{sieh-} \rangle$	$\langle du, -t \rangle$
				$\langle \text{dass}, \text{wen} \rangle$		$\langle \text{dass}, \text{Hans} \rangle$	$\langle \text{dass}, \text{sieh-} \rangle$	$\langle \text{dass}, -t \rangle$
				$\langle \text{Hans}, \text{wen} \rangle$			$\langle \text{Hans}, \text{sieh-} \rangle$	$\langle \text{Hans}, -t \rangle$
				$\langle \text{sieh-}, \text{wen} \rangle$			$\langle \text{sieh-}, \text{sieh-} \rangle$	$\langle \text{sieh-}, -t \rangle$
				$\langle -t, \text{wen} \rangle$			$\langle -t, \text{sieh-} \rangle$	

Table 2: The unrepaired s-precedence relation for (15). Grayed-out node labels are used to indicate null morphemes. Underlining indicates which ordered pairs must be deleted to repair the s-precedence relation.

$\langle \text{wen}, \text{wen} \rangle$	$\langle \text{wen}, \text{glaub-} \rangle$	$\langle \text{wen}, -st \rangle$	$\langle \text{wen}, C_1 \rangle$	$\langle \text{wen}, du \rangle$	$\langle \text{wen}, \text{dass} \rangle$	$\langle \text{wen}, \text{Hans} \rangle$	$\langle \text{wen}, \text{sieh-} \rangle$	$\langle \text{wen}, -t \rangle$
	$\langle \text{glaub-}, \text{glaub-} \rangle$	$\langle \text{glaub-}, -st \rangle$	$\langle \text{glaub-}, C_1 \rangle$	$\langle \text{glaub-}, du \rangle$	$\langle \text{glaub-}, \text{dass} \rangle$	$\langle \text{glaub-}, \text{Hans} \rangle$	$\langle \text{glaub-}, \text{sieh-} \rangle$	$\langle \text{glaub-}, -t \rangle$
		$\langle -st, -st \rangle$	$\langle -st, C_1 \rangle$	$\langle -st, du \rangle$	$\langle -st, \text{dass} \rangle$	$\langle -st, \text{Hans} \rangle$	$\langle -st, \text{sieh-} \rangle$	$\langle -st, -t \rangle$
				$\langle C_1, du \rangle$	$\langle C_1, \text{dass} \rangle$	$\langle C_1, \text{Hans} \rangle$	$\langle C_1, \text{sieh-} \rangle$	$\langle C_1, -t \rangle$
					$\langle du, \text{dass} \rangle$	$\langle du, \text{Hans} \rangle$	$\langle du, \text{sieh-} \rangle$	$\langle du, -t \rangle$
						$\langle \text{dass}, \text{Hans} \rangle$	$\langle \text{dass}, \text{sieh-} \rangle$	$\langle \text{dass}, -t \rangle$
							$\langle \text{Hans}, \text{sieh-} \rangle$	$\langle \text{Hans}, -t \rangle$
							$\langle \text{sieh-}, \text{sieh-} \rangle$	$\langle \text{sieh-}, -t \rangle$

Table 3: The repaired s-precedence relation for (15). Grayed-out node labels are used to indicate null morphemes.

Now, let us compare this to the linearization of (14). In this case, Vocabulary Insertion follows the mapping in (17), mapping the u-precedence relation in Table 1 onto the s-precedence relation in Table 4. For expositional purposes, we use indices to distinguish exponents.

(17) Vocabulary mapping for (14).

- a. $DP_1 \rightarrow \textit{wen}_1$
- b. $DP_1 \rightarrow \textit{wen}_2$
- c. $V_1 \rightarrow \textit{glaub-}$
- d. $T_1 \rightarrow \textit{-t}$
- e. $C_1 \rightarrow \emptyset$
- f. $DP_2 \rightarrow \textit{du}$
- g. $C_2 \rightarrow \emptyset$
- h. $DP_3 \rightarrow \textit{Hans}$
- i. $V_2 \rightarrow \textit{sieh-}$
- j. $T_2 \rightarrow \textit{-t}$

As with the previous example, this s-precedence relation is not antisymmetric, containing symmetric counterparts such as $\langle \textit{wen}_1, \textit{du} \rangle$ and $\langle \textit{du}, \textit{wen}_1 \rangle$, and thus requires repair. Deleting the underlined ordered pairs produces Table 5, which corresponds to the surface form in (14).¹²

$\langle \text{wen}_1, \text{glaub-} \rangle$	$\langle \text{wen}_1, \text{-st} \rangle$	$\langle \text{wen}_1, \text{C}_1 \rangle$	$\langle \text{wen}_1, \text{du} \rangle$	$\langle \text{wen}_1, \text{wen}_1 \rangle$	$\langle \text{wen}_1, \text{wen}_2 \rangle$	$\langle \text{wen}_1, \text{C}_2 \rangle$	$\langle \text{wen}_1, \text{Hans} \rangle$	$\langle \text{wen}_1, \text{sieh-} \rangle$	$\langle \text{wen}_1, \text{-t} \rangle$
$\langle \text{wen}_2, \text{glaub-} \rangle$	$\langle \text{wen}_2, \text{-st} \rangle$	$\langle \text{wen}_2, \text{C}_1 \rangle$	$\langle \text{wen}_2, \text{du} \rangle$	$\langle \text{wen}_2, \text{wen}_1 \rangle$	$\langle \text{wen}_2, \text{wen}_2 \rangle$	$\langle \text{wen}_2, \text{C}_2 \rangle$	$\langle \text{wen}_2, \text{Hans} \rangle$	$\langle \text{wen}_2, \text{sieh-} \rangle$	$\langle \text{wen}_2, \text{-t} \rangle$
$\langle \text{glaub-}, \text{glaub-} \rangle$	$\langle \text{glaub-}, \text{-st} \rangle$	$\langle \text{glaub-}, \text{C}_1 \rangle$	$\langle \text{glaub-}, \text{du} \rangle$	$\langle \text{glaub-}, \text{wen}_1 \rangle$	$\langle \text{glaub-}, \text{wen}_2 \rangle$	$\langle \text{glaub-}, \text{C}_2 \rangle$	$\langle \text{glaub-}, \text{Hans} \rangle$	$\langle \text{glaub-}, \text{sieh-} \rangle$	$\langle \text{glaub-}, \text{-t} \rangle$
$\langle \text{-st}, \text{glaub-} \rangle$	$\langle \text{-st}, \text{-st} \rangle$	$\langle \text{-st}, \text{C}_1 \rangle$	$\langle \text{-st}, \text{du} \rangle$	$\langle \text{-st}, \text{wen}_1 \rangle$	$\langle \text{-st}, \text{wen}_2 \rangle$	$\langle \text{-st}, \text{C}_2 \rangle$	$\langle \text{-st}, \text{Hans} \rangle$	$\langle \text{-st}, \text{sieh-} \rangle$	$\langle \text{-st}, \text{-t} \rangle$
$\langle \text{C}_1, \text{glaub-} \rangle$	$\langle \text{C}_1, \text{-st} \rangle$		$\langle \text{C}_1, \text{du} \rangle$	$\langle \text{C}_1, \text{wen}_1 \rangle$	$\langle \text{C}_1, \text{wen}_2 \rangle$	$\langle \text{C}_1, \text{C}_2 \rangle$	$\langle \text{C}_1, \text{Hans} \rangle$	$\langle \text{C}_1, \text{sieh-} \rangle$	$\langle \text{C}_1, \text{-t} \rangle$
$\langle \text{du}, \text{glaub-} \rangle$	$\langle \text{du}, \text{-st} \rangle$			$\langle \text{du}, \text{wen}_1 \rangle$	$\langle \text{du}, \text{wen}_2 \rangle$	$\langle \text{du}, \text{C}_2 \rangle$	$\langle \text{du}, \text{Hans} \rangle$	$\langle \text{du}, \text{sieh-} \rangle$	$\langle \text{du}, \text{-t} \rangle$
				$\langle \text{C}_2, \text{wen}_1 \rangle$	$\langle \text{C}_2, \text{wen}_2 \rangle$		$\langle \text{C}_2, \text{Hans} \rangle$	$\langle \text{C}_2, \text{sieh-} \rangle$	$\langle \text{C}_2, \text{-t} \rangle$
				$\langle \text{Hans}, \text{wen}_1 \rangle$	$\langle \text{Hans}, \text{wen}_2 \rangle$			$\langle \text{Hans}, \text{sieh-} \rangle$	$\langle \text{Hans}, \text{-t} \rangle$
				$\langle \text{sieh-}, \text{wen}_1 \rangle$	$\langle \text{sieh-}, \text{wen}_2 \rangle$			$\langle \text{sieh-}, \text{sieh-} \rangle$	$\langle \text{sieh-}, \text{-t} \rangle$
				$\langle \text{-t}, \text{wen}_1 \rangle$	$\langle \text{-t}, \text{wen}_2 \rangle$			$\langle \text{-t}, \text{sieh-} \rangle$	

Table 4: The unrepaired s-precedence relation for (14). Grayed-out node labels are used to indicate null morphemes. Underlining indicates which ordered pairs must be deleted to repair the s-precedence relation.

$\langle \text{wen}_1, \text{wen}_1 \rangle$	$\langle \text{wen}_1, \text{glaub-} \rangle$	$\langle \text{wen}_1, \text{-st} \rangle$	$\langle \text{wen}_1, \text{C}_1 \rangle$	$\langle \text{wen}_1, \text{du} \rangle$	$\langle \text{wen}_1, \text{wen}_2 \rangle$	$\langle \text{wen}_1, \text{C}_2 \rangle$	$\langle \text{wen}_1, \text{Hans} \rangle$	$\langle \text{wen}_1, \text{sieh-} \rangle$	$\langle \text{wen}_1, \text{-t} \rangle$
	$\langle \text{glaub-}, \text{glaub-} \rangle$	$\langle \text{glaub-}, \text{-st} \rangle$	$\langle \text{glaub-}, \text{C}_1 \rangle$	$\langle \text{glaub-}, \text{du} \rangle$	$\langle \text{glaub-}, \text{wen}_2 \rangle$	$\langle \text{glaub-}, \text{C}_2 \rangle$	$\langle \text{glaub-}, \text{Hans} \rangle$	$\langle \text{glaub-}, \text{sieh-} \rangle$	$\langle \text{glaub-}, \text{-t} \rangle$
		$\langle \text{-st}, \text{-st} \rangle$	$\langle \text{-st}, \text{C}_1 \rangle$	$\langle \text{-st}, \text{du} \rangle$	$\langle \text{-st}, \text{wen}_2 \rangle$	$\langle \text{-st}, \text{C}_2 \rangle$	$\langle \text{-st}, \text{Hans} \rangle$	$\langle \text{-st}, \text{sieh-} \rangle$	$\langle \text{-st}, \text{-t} \rangle$
				$\langle \text{C}_1, \text{du} \rangle$	$\langle \text{C}_1, \text{wen}_2 \rangle$	$\langle \text{C}_1, \text{C}_2 \rangle$	$\langle \text{C}_1, \text{Hans} \rangle$	$\langle \text{C}_1, \text{sieh-} \rangle$	$\langle \text{C}_1, \text{-t} \rangle$
					$\langle \text{du}, \text{wen}_2 \rangle$	$\langle \text{du}, \text{C}_2 \rangle$	$\langle \text{du}, \text{Hans} \rangle$	$\langle \text{du}, \text{sieh-} \rangle$	$\langle \text{du}, \text{-t} \rangle$
					$\langle \text{wen}_2, \text{wen}_2 \rangle$	$\langle \text{wen}_2, \text{C}_2 \rangle$	$\langle \text{wen}_2, \text{Hans} \rangle$	$\langle \text{wen}_2, \text{sieh-} \rangle$	$\langle \text{wen}_2, \text{-t} \rangle$
							$\langle \text{C}_2, \text{Hans} \rangle$	$\langle \text{C}_2, \text{sieh-} \rangle$	$\langle \text{C}_2, \text{-t} \rangle$
								$\langle \text{Hans}, \text{sieh-} \rangle$	$\langle \text{Hans}, \text{-t} \rangle$
								$\langle \text{sieh-}, \text{sieh-} \rangle$	$\langle \text{sieh-}, \text{-t} \rangle$

Table 5: The repaired s-precedence relation for (14). Grayed-out node labels are used to indicate null morphemes.

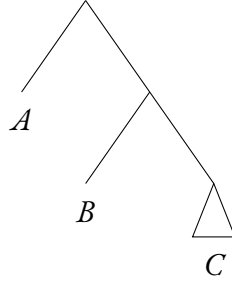


Figure 2: An abstract tree.

Note that fourteen ordered pairs were deleted in generating (15), while twenty-three were deleted in producing (14). In general, because multiple exponence introduces more ordered pairs into the s-precedence relation, it requires more ordered pairs to be deleted. To the extent that (i) the repair approach is correct and (ii) deletions of ordered pairs are subject to economy considerations, the increased cost of multiple exponence may contribute to the cross-linguistic preference for single exponence.

Additionally, note that in (14), the node that is realized twice has moved. In fact, movement is required to generate fully superfluous multiple exponence. To illustrate, consider the simple tree in Figure 2, which linearizes into the u-precedence relation $P_u = \{\langle A, B \rangle, \langle A, C \rangle, \langle B, C \rangle\}$. If Vocabulary Insertion inserts some a_1, a_2 for A , b for B , and c for C , this then gives the s-precedence relation $P_s = \{\langle a_1, b \rangle, \langle a_1, c \rangle, \langle a_2, b \rangle, \langle a_2, c \rangle, \langle b, c \rangle\}$. Here, there is a violation not of antisymmetry, but of totality: the two exponents of A , a_1 and a_2 , are not ordered with respect to each other. Because there is no mechanism for adding ordered pairs to the s-precedence relation, we cannot repair this violation, so linearization fails. Thus, fully superfluous multiple exponence requires the u-precedence relation to contain reflexive ordered pairs. If displacement is the only source of reflexivity in linearization, then this predicts that fully superfluous multiple exponence can only arise when displacement has occurred.

As shown in this section, by allowing Vocabulary Insertion to insert multiple exponents for a single node, we can produce multiple exponence. This account does not depend on any

postsyntactic operations other than Vocabulary Insertion and linearization—there is no need for Fission or Fusion. Furthermore, the present account does not preclude the possibility that apparent multiple exponence may arise in other ways as well.¹³ For example, some cases of apparent multiple exponence may involve multiple projections in the syntax, as argued by Vanden Wyngaerd et al. (2021), while others may result from multiple agreement (e.g., Carstens, 2011). Additionally, unlike Nunes’s (2004) account of “multiple copy spell-out” (see footnote 3), our account of multiple exponence does not assume the existence of copies, and thus is compatible with the multidominance theory of movement. Moreover, this proposal can account for multiple exponence involving portmanteaux. We demonstrate this in section 3.5, after first introducing our theory of portmanteaux.

3.3 Simultaneity

Simultaneity, the phenomenon in which multiple meaningful elements are realized at the same time, abounds in natural language (Kremers, 2012). For example, (18) shows simultaneity in Ekhwa Adara (Niger-Congo: Nigeria; based on our fieldnotes) between the plural morpheme /H/ and the noun stem, a type of simultaneity that is typical of languages with grammatical tone (Lionnet et al., 2022). Similarly, (19) demonstrates simultaneity in German Sign Language between the non-manual negation marker and the verb phrase [VP FLOWER BUY]. Such simultaneity between manual and non-manual signs is common in signed languages (Kremers, 2012).

(18)	Singular	Plural	Gloss
	<i>enéhíne</i>	<i>énéhíne</i>	‘woman’
	<i>ete</i>	<i>éte</i>	‘father’
	<i>ewé</i>	<i>éwé</i>	‘child’
	<i>avò-ŋ’</i>	<i>ávò-ŋ’</i>	‘goat’

(19) MOTHER $\overline{\text{FLOWER BUY}}^{\text{neg}}$

‘Mother does not buy a flower’ (Pfau & Quer, 2002, 75)

It is typically assumed that linearization produces sequential (i.e., non-simultaneous) representations. This entails that any surface simultaneity arises later due to phonological or phonetic processes. Conceptually, however, there is no clear reason that simultaneity should be banned at linearization. If anything, it should be expected: if we accept that linearization is an intermodular (PF) operation (Chomsky, 1995), then its role is to help translate syntactic representations into phonological ones. Given that the phonology clearly tolerates simultaneity, we would expect that linearization tolerates it as well, since the goal of linearization is to help produce a phonological representation. In other words, linearization should not produce proper precedence when phonology appears to operate on non-proper precedence.

In general, asymmetric models of linearization generate sequential representations and are therefore unable to account for simultaneity without further assumptions. In (18), the high tone of the plural morpheme is associated with the first syllable, so that we have [enénjine] ‘woman’ but [énénjine] ‘women’, for example. Because asymmetry entails irreflexivity, it implies the irreflexive relation of proper precedence, repeated below.

(13) Given lexical items x, y , x *properly precedes* y ($x < y$) if x is pronounced before y .

However, neither /H/ nor /enénjine/ properly precedes the other: they are pronounced at the same time. Consequently, if we adopt asymmetry (and thus proper precedence), we end up with a failure of totality: a floating morpheme is not ordered with respect to its host (see Kremers, 2012, 2013). The German Sign Language example poses a similar challenge: the non-manual marker neither properly precedes nor is properly preceded by FLOWER and BUY, so there is a failure of totality under asymmetry. An antisymmetric proposal can avoid such problems. To briefly illustrate, suppose that we have a u-precedence relation P_u with $\langle \text{Neg}, \text{N} \rangle, \langle \text{Neg}, \text{V} \rangle, \langle \text{N}, \text{V} \rangle \in$

P_u , and that Vocabulary Insertion inserts neg for Neg, FLOWER for N, and BUY for V. This gives an s-precedence relation P_s with $\langle \text{neg}, \text{FLOWER} \rangle, \langle \text{neg}, \text{BUY} \rangle, \langle \text{FLOWER}, \text{BUY} \rangle \in P_s$. Then this can be realized as in (19), with FLOWER properly preceding BUY, while the non-manual neg is pronounced at the same time as both. It is clear that this pronunciation satisfies the Precedence Mapping Principle for $\langle \text{FLOWER}, \text{BUY} \rangle$. The Precedence Mapping Principle is satisfied for $\langle \text{neg}, \text{FLOWER} \rangle$ as well, because neg improperly precedes FLOWER, and it is satisfied for $\langle \text{neg}, \text{BUY} \rangle$ for the same reason. It is not a problem that FLOWER ends before neg does, because the converse of the Precedence Mapping Principle does not hold, as discussed in section 3.1. In other words, even if FLOWER in some sense precedes part of neg, it does not follow that $\langle \text{FLOWER}, \text{neg} \rangle \in P_s$.

In addition to the conceptual argument for allowing simultaneity in linearization, there is empirical evidence to suggest that in certain cases, simultaneity must arise at PF. Nupe (Niger-Congo: Nigeria) has a polarity focus construction in which the verb is realized twice, as in (20).

- (20) Musa ba nakàn ba
Musa cut meat cut
‘Musa DID cut the meat’ (Kandybowicz, 2007, 136)

According to Kandybowicz (2006, 2007, 2008), the exponent of the focus head is a floating low tone /L/. Because this tonal morpheme needs a host, it triggers insertion of an additional exponent of the verb (in Kandybowicz’s framework, spell-out of a lower copy). If this is indeed the case, then the linking between /L/ ‘FOC’ and *ba* ‘cut’ cannot happen in the phonology. We demonstrate this via a proof by contradiction. Suppose that simultaneity only arises in phonology, with the corollary that linearization produces sequential representations. Then /L/ ‘FOC’ does not need—indeed, cannot have—a host at PF, because there is no simultaneity between exponents at this stage. Because /L/ ‘FOC’ does not yet need a host, there is nothing to trigger insertion of a second instance of *ba* ‘cut’.¹⁴ We thus incorrectly predict the surface form *Musa ba nakàn*, with only one instance of *ba* ‘cut’. Thus, the simultaneity between /L/ ‘FOC’ and *ba*

‘cut’ cannot arise in the phonology (by process of elimination, it must arise at PF). Therefore, it is not the case that simultaneity only arises in phonology. Tonological interactions further support the intermodular analysis of the Nupe polarity focus construction. Instead of forming a contour with the lexical tone of the verbal host—the expected outcome if the tonal morpheme docks onto the verb in the phonology—/L/ ‘FOC’ triggers a leftwards “downstep” not otherwise attested in Nupe phonology, suggesting that /L/ ‘FOC’ does not link to its host via a phonological process (Kandybowicz, 2004). In conclusion, the Nupe data provide evidence for simultaneity arising at PF rather than in the phonology.

If the simultaneity between *ba* ‘cut’ and the tonal focus marker arises at PF—that is, during linearization—then this poses a challenge for asymmetric models of linearization, as discussed above for Ekhwa Adara and German Sign Language. Specifically, under an asymmetric, sequential model, *ba* ‘cut’ and /L/ ‘FOC’ are not ordered with respect to each other (they are not in a relationship of proper precedence), so there is a failure of totality. In contrast, this example poses no issue if linear order is antisymmetric. As discussed in section 3.1, an antisymmetric definition of linear order entails non-proper precedence. Under this conception of precedence, *ba* ‘cut’ precedes /L/ ‘FOC’ because they are pronounced at the same time, and /L/ ‘FOC’ precedes *ba* for the same reason.

To illustrate, let us walk through the linearization of (20). We assume the structure in Figure 3, adapted from Kandybowicz (2007, (28)). Indices indicate syntactic identity. Following Kandybowicz (2007), we assume that linearization produces the u-precedence relation in Table 6 and that Vocabulary Insertion produces the mapping in (21), producing the s-precedence relation in Table 7. Note that V has two phonologically identical exponents, with indices used to distinguish them.

(21) Vocabulary mapping for (20).

a. $DP_i \rightarrow Musa$

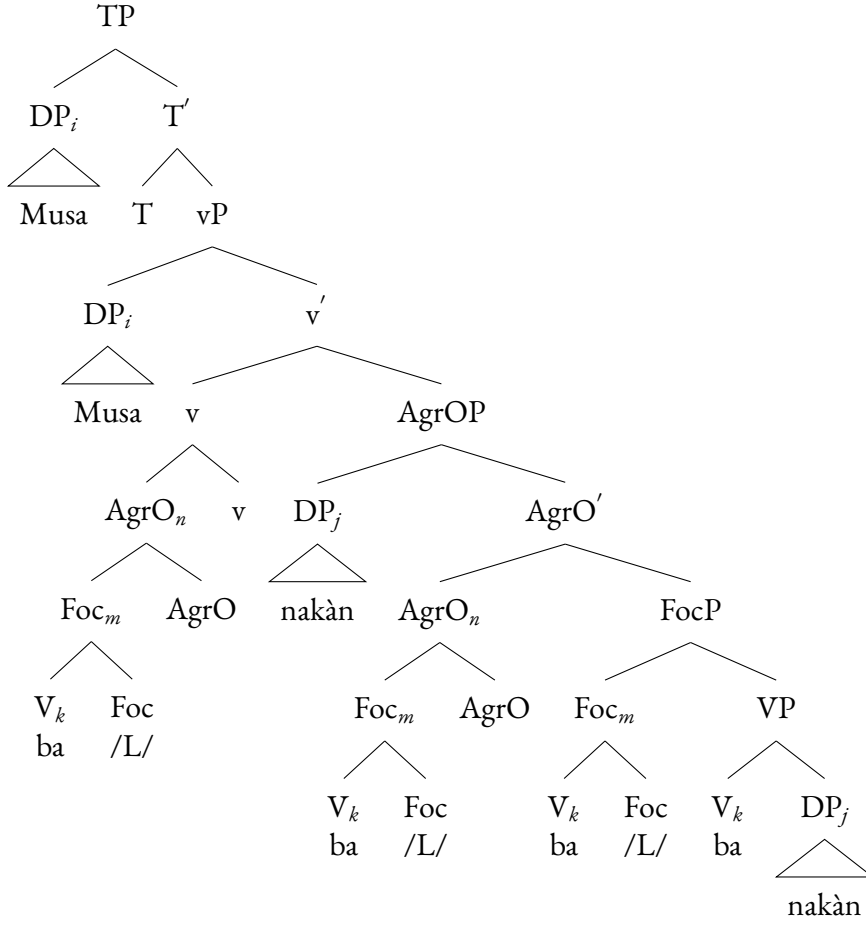


Figure 3: Syntactic structure for (20).

{	$\langle DP_i, T \rangle$	$\langle DP_i, DP_i \rangle$	$\langle DP_i, V \rangle$	$\langle DP_i, Foc \rangle$	$\langle DP_i, AgrO \rangle$	$\langle DP_i, v \rangle$	$\langle DP_i, DP_j \rangle$
		$\langle T, DP_i \rangle$	$\langle T, V \rangle$	$\langle T, Foc \rangle$	$\langle T, AgrO \rangle$	$\langle T, v \rangle$	$\langle T, DP_j \rangle$
			$\langle V, V \rangle$	$\langle V, Foc \rangle$	$\langle V, AgrO \rangle$	$\langle V, v \rangle$	$\langle V, DP_j \rangle$
			$\langle Foc, V \rangle$	$\langle Foc, Foc \rangle$	$\langle Foc, AgrO \rangle$	$\langle Foc, v \rangle$	$\langle Foc, DP_j \rangle$
			$\langle AgrO, V \rangle$	$\langle AgrO, Foc \rangle$	$\langle AgrO, AgrO \rangle$	$\langle AgrO, v \rangle$	$\langle AgrO, DP_j \rangle$
			$\langle v, V \rangle$	$\langle v, Foc \rangle$	$\langle v, AgrO \rangle$	$\langle v, v \rangle$	$\langle v, DP_j \rangle$
			$\langle DP_j, V \rangle$	$\langle DP_j, Foc \rangle$	$\langle DP_j, AgrO \rangle$	$\langle DP_j, v \rangle$	$\langle DP_j, DP_j \rangle$

Table 6: The u-precedence relation for Figure 3. For expositional purposes, we ignore the internal structure of the DPs.

- b. $T \rightarrow \emptyset$
- c. $V \rightarrow ba_1$
- d. $V \rightarrow ba_2$
- e. $\text{Foc} \rightarrow /L/$
- f. $\text{AgrO} \rightarrow \emptyset$
- g. $v \rightarrow \emptyset$
- h. $\text{DP}_j \rightarrow \textit{nakàn}$

In order to produce the surface form, the underlined ordered pairs in Table 7 must be deleted.¹⁵ This produces the s-precedence relation in Table 8. The rows and columns have been reordered to make the surface order clearer. This s-precedence relation maps onto the surface string: since ba_2 is the only exponent adjacent to $/L/$, it will necessarily serve as its host. This demonstrates how simultaneity arises under our proposal.

$$\left\{ \begin{array}{cccccccc} \langle Musa, T \rangle & \langle Musa, Musa \rangle & \langle Musa, ba_1 \rangle & \langle Musa, ba_2 \rangle & \langle Musa, /L/ \rangle & \langle Musa, AgrO \rangle & \langle Musa, v \rangle & \langle Musa, nakàn \rangle \\ & \underline{\langle T, Musa \rangle} & \langle T, ba_1 \rangle & \langle T, ba_1 \rangle & \langle T, /L/ \rangle & \langle T, AgrO \rangle & \langle T, v \rangle & \langle T, nakàn \rangle \\ & & \langle ba_1, ba_1 \rangle & \langle ba_1, ba_2 \rangle & \langle ba_1, /L/ \rangle & \langle ba_1, AgrO \rangle & \langle ba_1, v \rangle & \langle ba_1, nakàn \rangle \\ & & \underline{\langle ba_2, ba_1 \rangle} & \langle ba_2, ba_2 \rangle & \langle ba_2, /L/ \rangle & \underline{\langle ba_2, AgrO \rangle} & \underline{\langle ba_2, v \rangle} & \underline{\langle ba_2, nakàn \rangle} \\ & & \underline{\langle /L/, ba_1 \rangle} & \underline{\langle /L/, ba_2 \rangle} & \langle /L/, /L/ \rangle & \underline{\langle /L/, AgrO \rangle} & \underline{\langle /L/, v \rangle} & \underline{\langle /L/, nakàn \rangle} \\ & & \underline{\langle AgrO, ba_1 \rangle} & \underline{\langle AgrO, ba_2 \rangle} & \langle AgrO, /L/ \rangle & \underline{\langle AgrO, AgrO \rangle} & \underline{\langle AgrO, v \rangle} & \underline{\langle AgrO, nakàn \rangle} \\ & & \underline{\langle v, ba_1 \rangle} & \langle v, ba_2 \rangle & \langle v, /L/ \rangle & \underline{\langle v, AgrO \rangle} & \langle v, v \rangle & \langle v, nakàn \rangle \\ & & \underline{\langle nakàn, ba_1 \rangle} & \langle nakàn, ba_2 \rangle & \langle nakàn, /L/ \rangle & \underline{\langle nakàn, AgrO \rangle} & \underline{\langle nakàn, v \rangle} & \langle nakàn, nakàn \rangle \end{array} \right\}$$

Table 7: The unrepaired s-precedence relation for Figure 3. Grayed-out node labels are used to indicate null morphemes.

$$\left\{ \begin{array}{cccccccc} \langle Musa, T \rangle & \langle Musa, Musa \rangle & \langle Musa, ba_1 \rangle & \langle Musa, AgrO \rangle & \langle Musa, v \rangle & \langle Musa, nakàn \rangle & \langle Musa, ba_2 \rangle & \langle Musa, /L/ \rangle \\ & & \langle T, ba_1 \rangle & \langle T, AgrO \rangle & \langle T, v \rangle & \langle T, nakàn \rangle & \langle T, ba_1 \rangle & \langle T, /L/ \rangle \\ & & \langle ba_1, ba_1 \rangle & \langle ba_1, AgrO \rangle & \langle ba_1, v \rangle & \langle ba_1, nakàn \rangle & \langle ba_1, ba_2 \rangle & \langle ba_1, /L/ \rangle \\ & & & \langle AgrO, AgrO \rangle & \langle AgrO, v \rangle & \langle AgrO, nakàn \rangle & \langle AgrO, ba_2 \rangle & \langle AgrO, /L/ \rangle \\ & & & & \langle v, v \rangle & \langle v, nakàn \rangle & \langle v, ba_2 \rangle & \langle v, /L/ \rangle \\ & & & & & \langle nakàn, nakàn \rangle & \langle nakàn, ba_2 \rangle & \langle nakàn, /L/ \rangle \\ & & & & & & \langle ba_2, ba_2 \rangle & \langle ba_2, /L/ \rangle \\ & & & & & & & \langle /L/, /L/ \rangle \end{array} \right\}$$

Table 8: The repaired s-precedence relation for Figure 3. Grayed-out node labels are used to indicate null morphemes.

To conclude, this section has presented evidence from Nupe that certain instances of simultaneity arise at PF rather than in the phonology. This is problematic for an asymmetric conception of linear order, where simultaneity precludes totality—there is no proper precedence between simultaneous elements—but is compatible with an antisymmetric approach.

3.4 Portmanteaux

Because it treats linear order as antisymmetric and treats Vocabulary Insertion as many-to-many, the current proposal allows for the insertion of portmanteaux (or cumulative morphemes), morphemes that realize multiple grammatical categories. The conceptual insight is that portmanteaux are an example not only of a many-to-one mapping (see section 2.2), but also of simultaneity: when multiple nodes are realized by a single exponent, they are realized simultaneously.

Portmanteaux can be inserted under two structural configurations under our proposal. First, a portmanteau can be inserted to realize linearly adjacent nodes; this has been argued independently by various authors (e.g., Williams, 2003; Johnson, 2012; Woolford, 2016; Ostrove, 2018). Second, a portmanteau can be inserted when one node has crossed over another. To our knowledge, the prediction of portmanteaux based on crossover is novel to our proposal.

3.4.1 Adjacency-based portmanteaux

Let us first discuss adjacency-based portmanteaux, using the example of Irish dependent verb forms. Certain Irish verbs display a suppletive alternation between “independent” and “dependent” forms, shown in Table 9, with the dependent forms appearing following certain complementizers.

Following Oda (2012), Ostrove (2018) argues that dependent verbs are portmanteaux which expone the root, *v*, and Mood, since dependent verbs are in complementary distribution with overt exponence of Mood. However, Mood is taken to occupy a high position between

Verb	Gloss	Tense	Independent	Dependent
<i>bí</i>	‘be’	past	<i>bhí</i>	<i>raibh</i>
		present	<i>tá</i>	<i>fuil</i>
<i>déan</i>	‘do’	past	<i>rinne</i>	<i>dearna</i>
<i>faigh</i>	‘get’	past	<i>fuair</i>	<i>bhfuair</i>
		future/conditional	<i>gheobhaidh</i>	<i>bhfaighidh</i>
<i>feic</i>	‘see’	past	<i>chonaic</i>	<i>faca</i>
		present/future	<i>chí/tí</i>	<i>feic</i>
<i>téigh</i>	‘go’	past	<i>chuaigh</i>	<i>deachaigh</i>

Table 9: The dependent alternation in Irish verbs (Ostrove, 2018, (23)).

T and C, so $\sqrt{\cdot}$, v , and Mood do not constitute a span (see Figure 4 below). Because of this, Ostrove proposes the stretch, a sequence of linearly adjacent nodes, as a target for Vocabulary Insertion. We will use the same data to show how portmanteau are inserted under adjacency under our proposal.

Ostrove analyzes the example in (22), for which they adopt the structure in Figure 4.

- (22) An bhfac-aís an cailín úd?
Q see.PAST.DEP-PAST.2SG the girl yonder
‘Did you see that girl over yonder?’ (Sayers, 2009, 56; in Ostrove, 2018, (6b))

Following Ostrove (2018), linearizing the structure in Figure 4 will produce the u-precedence relation in Table 10. Assuming that Vocabulary Insertion inserts the exponents in (23), the u-precedence relation is mapped onto the s-precedence relation in Table 11.¹⁶ We use indices to indicate that Mood, $\sqrt{\text{SEE}}$, and v are mapped onto the same exponent *bhfac-*, rather than to distinct but identical exponents.

$$\left\{ \begin{array}{cccc} \langle C, \text{Mood} \rangle & \langle C, \sqrt{\text{SEE}} \rangle & \langle C, v \rangle & \langle C, T \rangle \\ & \langle \text{Mood}, \sqrt{\text{SEE}} \rangle & \langle \text{Mood}, v \rangle & \langle \text{Mood}, T \rangle \\ & & \langle \sqrt{\text{SEE}}, v \rangle & \langle \sqrt{\text{SEE}}, T \rangle \\ & & & \langle v, T \rangle \end{array} \right\}$$

Table 10: The u-precedence relation for the structure in Figure 4.

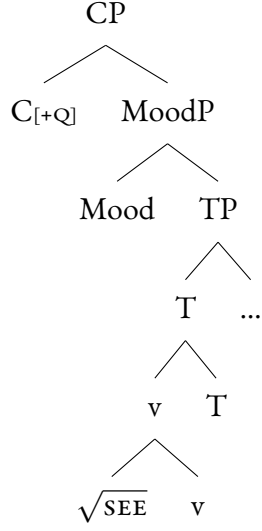


Figure 4: The structure underlying (22) (Ostrove, 2018, (49)).

(23) Vocabulary mapping for (22).

- a. $C \rightarrow an$
- b. $Mood \rightarrow bhfac_{-1}$
- c. $\sqrt{SEE} \rightarrow bhfac_{-1}$
- d. $v \rightarrow bhfac_{-1}$
- e. $T \rightarrow -aís$

$$\left\{ \begin{array}{cccc} \langle an, bhfac \rangle & \langle an, bhfac \rangle & \langle an, bhfac \rangle & \langle an, -aís \rangle \\ & \langle bhfac, bhfac \rangle & \langle bhfac, bhfac \rangle & \langle bhfac, -aís \rangle \\ & & \langle bhfac, bhfac \rangle & \langle bhfac, -aís \rangle \\ & & & \langle bhfac, -aís \rangle \end{array} \right\}$$

Table 11: The unsimplified s-precedence relation for the structure in Figure 4.

Once we remove repetitions, which have no status in sets, Table 11 simplifies to Table 12, which requires no repair. This s-precedence relation will be mapped onto the observed string in (22).

The preceding discussion shows that adjacency is sufficient to insert a portmanteau in this example, but not that it is necessary. To demonstrate the latter point, consider the hypotheti-

$$\left\{ \begin{array}{ll} \langle an, bhfac- \rangle & \langle an, -ais \rangle \\ \langle bhfac-, bhfac- \rangle & \langle bhfac-, -ais \rangle \end{array} \right\}$$

Table 12: The simplified s-precedence relation for the structure in Figure 4.

cal structure in Figure 5, where a projection XP appears between MoodP and TP. In this case,

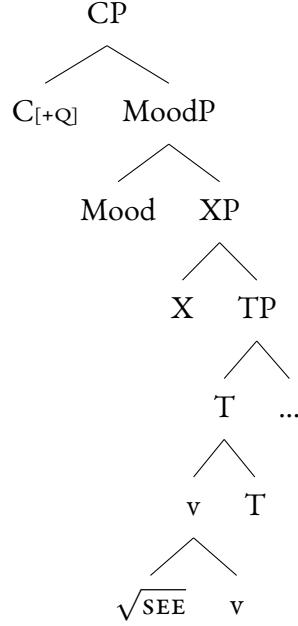


Figure 5: A hypothetical structure for (22).

linearization produces the u-precedence relation in Table 13. If we insert some exponent x for X , otherwise preserving the mappings in (23), then the u-precedence relation is mapped onto the s-precedence relation in Table 14, which simplifies to Table 15 after removing repetitions. In this case, the s-precedence relation is not antisymmetric, containing both $\langle bhfac-, x \rangle$ and $\langle x, bhfac- \rangle$. Moreover, the Principle of Maximality prevents this violation of antisymmetry from being resolved. The pair $\langle bhfac-, x \rangle$ realizes $\langle \text{Mood}, X \rangle$ (among other pairs), which has no counterpart $\langle X, \text{Mood} \rangle$, so $\langle bhfac-, x \rangle$ cannot be deleted. Similarly, $\langle x, bhfac- \rangle$ realizes $\langle X, v \rangle$ (among other pairs), which has no counterpart $\langle v, X \rangle$, so $\langle x, bhfac- \rangle$ cannot be deleted either. Thus, there is no way to repair the s-precedence relation, so externalization fails. This

$$\left\{ \begin{array}{ccccc} \langle C, \text{Mood} \rangle & \langle C, X \rangle & \langle C, \sqrt{\text{SEE}} \rangle & \langle C, v \rangle & \langle C, T \rangle \\ & \langle \text{Mood}, X \rangle & \langle \text{Mood}, \sqrt{\text{SEE}} \rangle & \langle \text{Mood}, v \rangle & \langle \text{Mood}, T \rangle \\ & & \langle X, \sqrt{\text{SEE}} \rangle & \langle X, v \rangle & \langle X, T \rangle \\ & & \langle \sqrt{\text{SEE}}, v \rangle & \langle \sqrt{\text{SEE}}, T \rangle & \\ & & & \langle v, T \rangle & \end{array} \right\}$$

Table 13: The u-precedence relation for the structure in Figure 5.

$$\left\{ \begin{array}{ccccc} \langle an, bhfac- \rangle & \langle an, x \rangle & \langle an, bhfac- \rangle & \langle an, bhfac- \rangle & \langle an, -ais \rangle \\ & \langle bhfac-, x \rangle & \langle bhfac-, bhfac- \rangle & \langle bhfac-, bhfac- \rangle & \langle bhfac-, -ais \rangle \\ & & \langle x, bhfac- \rangle & \langle x, bhfac- \rangle & \langle x, -ais \rangle \\ & & \langle bhfac-, bhfac- \rangle & \langle bhfac-, -ais \rangle & \\ & & & \langle bhfac-, -ais \rangle & \end{array} \right\}$$

Table 14: The unsimplified s-precedence relation for the structure in Figure 5.

$$\left\{ \begin{array}{ccc} \langle an, bhfac- \rangle & \langle an, x \rangle & \langle an, -ais \rangle \\ \langle bhfac-, bhfac- \rangle & \langle bhfac-, x \rangle & \langle bhfac-, -ais \rangle \\ \langle x, bhfac- \rangle & & \langle x, -ais \rangle \end{array} \right\}$$

Table 15: The simplified s-precedence relation for the structure in Figure 5.

demonstrates that adjacency is necessary for the insertion of the portmanteau *bhfac*-.: if adjacency is interrupted, then attempting to insert the portmanteau makes linearization impossible.¹⁷

3.4.2 Crossover-based portmanteaux

In addition to adjacency, movement of one node past another creates a configuration in which a portmanteau can be inserted, regardless of whether the nodes it realizes are adjacent. We demonstrate with negative suppletion in Korean. In Korean, *al*- ‘know’ has a suppletive form *molu*- used in certain negative constructions, specifically so-called “short-form” negation:

- (24) Halapeci-kkeyse Inho-lul molu-si-ess-ta
 grandfather-NOM.HON Inho-ACC know.NEG-HON-PST-DECL
 ‘Grandfather didn’t know Inho’ (Choi & Harley, 2019, (53a))

We assume for (24) the structure in Figure 6, with movement the subject from the Spec,vP to Spec,CP via Spec,TP and head movement of $\sqrt{\text{KNOW}}$ to C via the intermediate heads. We follow Choi & Harley (2019) in taking the honorific node Hon to occupy a lower position than Neg, *pace* Chung (2009).¹⁸ Because $\sqrt{}$ is not adjacent to Neg, these data are challenging for approaches to portmanteaux and suppletion that rely on structural adjacency. For example, because Hon is realized separately as *-si*, $\sqrt{}$ and Neg cannot form a span.¹⁹

We assume a specifier-initial, head-final linearization (see Choi & Harley, 2019, (27)). Thus, linearizing the structure in Figure 6 will generate the u-precedence relation in Table 16. Note that $\sqrt{\text{KNOW}}$, *v*, Neg, Hon, and T both precede and are preceded by each other. This is because they are not only dominated by TP but contained within the complex C head, which follows TP. We ignore the internal structure of the DP and PP, as it is irrelevant to the current discussion.

{	$\langle \text{DP}_s, \text{DP}_s \rangle$	$\langle \text{DP}_s, \text{DP}_o \rangle$	$\langle \text{DP}_s, \sqrt{\text{KNOW}} \rangle$	$\langle \text{DP}_s, v \rangle$	$\langle \text{DP}_s, \text{Hon} \rangle$	$\langle \text{DP}_s, \text{Neg} \rangle$	$\langle \text{DP}_s, \text{T} \rangle$	$\langle \text{DP}_s, \text{C} \rangle$
			$\langle \text{DP}_o, \sqrt{\text{KNOW}} \rangle$	$\langle \text{DP}_o, v \rangle$	$\langle \text{DP}_o, \text{Hon} \rangle$	$\langle \text{DP}_o, \text{Neg} \rangle$	$\langle \text{DP}_o, \text{T} \rangle$	$\langle \text{DP}_o, \text{C} \rangle$
			$\langle \sqrt{\text{KNOW}}, \sqrt{\text{KNOW}} \rangle$	$\langle \sqrt{\text{KNOW}}, v \rangle$	$\langle \sqrt{\text{KNOW}}, \text{Hon} \rangle$	$\langle \sqrt{\text{KNOW}}, \text{Neg} \rangle$	$\langle \sqrt{\text{KNOW}}, \text{T} \rangle$	$\langle \sqrt{\text{KNOW}}, \text{C} \rangle$
			$\langle v, \sqrt{\text{KNOW}} \rangle$	$\langle v, v \rangle$	$\langle v, \text{Hon} \rangle$	$\langle v, \text{Neg} \rangle$	$\langle v, \text{T} \rangle$	$\langle v, \text{C} \rangle$
			$\langle \text{Hon}, \sqrt{\text{KNOW}} \rangle$	$\langle \text{Hon}, v \rangle$	$\langle \text{Hon}, \text{Hon} \rangle$	$\langle \text{Hon}, \text{Neg} \rangle$	$\langle \text{Hon}, \text{T} \rangle$	$\langle \text{Hon}, \text{C} \rangle$
			$\langle \text{Neg}, \sqrt{\text{KNOW}} \rangle$	$\langle \text{Neg}, v \rangle$	$\langle \text{Neg}, \text{Hon} \rangle$	$\langle \text{Neg}, \text{Neg} \rangle$	$\langle \text{Neg}, \text{T} \rangle$	$\langle \text{Neg}, \text{C} \rangle$
			$\langle \text{T}, \sqrt{\text{KNOW}} \rangle$	$\langle \text{T}, v \rangle$	$\langle \text{T}, \text{Hon} \rangle$	$\langle \text{T}, \text{Neg} \rangle$	$\langle \text{T}, \text{T} \rangle$	$\langle \text{T}, \text{C} \rangle$

Table 16: The u-precedence relation for the structure in Figure 6.

Assuming that Vocabulary Insertion provides the exponents in (25), the u-precedence relation in Table 16 maps onto the s-precedence relation in Table 17, which simplifies to Table 18 after removing repetitions.

(25) Vocabulary mapping for (24).

- a. $DP_s \rightarrow \textit{halapeci-kkeyse}$
- b. $DP_o \rightarrow \textit{Inbo-lul}$
- c. $\sqrt{\text{KNOW}} \rightarrow \textit{molu}_{-1}$
- d. $v \rightarrow \emptyset$
- e. $\text{Hon} \rightarrow \textit{-si}$
- f. $\text{Neg} \rightarrow \textit{molu}_{-1}$
- g. $T \rightarrow \textit{-ess}$
- h. $C \rightarrow \textit{-ta}$

This s-precedence relation is not antisymmetric. For example, it contains both $\langle \textit{molu}_{-1}, \textit{-si} \rangle$ and $\langle \textit{-si}, \textit{molu}_{-1} \rangle$. To produce the observed linear order, the underlined ordered pairs must be deleted, producing the s-precedence relation in Table 19.

$$\left\{ \begin{array}{cccccccc} \langle \text{DP}_s, \text{DP}_s \rangle & \langle \text{DP}_s, \text{DP}_o \rangle & \langle \text{DP}_s, \text{molu-} \rangle & \langle \text{DP}_s, \emptyset \rangle & \langle \text{DP}_s, -si \rangle & \langle \text{DP}_s, \text{molu-} \rangle & \langle \text{DP}_s, -ess \rangle & \langle \text{DP}_s, -ta \rangle \\ \langle \text{DP}_o, \text{molu-} \rangle & \langle \text{DP}_o, \emptyset \rangle & \langle \text{DP}_o, -si \rangle & \langle \text{DP}_o, \text{molu-} \rangle & \langle \text{DP}_o, -ess \rangle & \langle \text{DP}_o, -ta \rangle & & \\ \langle \text{molu-}, \text{molu-} \rangle & \langle \text{molu-}, \emptyset \rangle & \langle \text{molu-}, -si \rangle & \langle \text{molu-}, \text{molu-} \rangle & \langle \text{molu-}, -ess \rangle & \langle \text{molu-}, -ta \rangle & & \\ \langle \emptyset, \text{molu-} \rangle & \langle \emptyset, \emptyset \rangle & \langle \emptyset, -si \rangle & \langle \emptyset, \text{molu-} \rangle & \langle \emptyset, -ess \rangle & \langle \emptyset, -ta \rangle & & \\ \langle -si, \text{molu-} \rangle & \langle -si, \emptyset \rangle & \langle -si, -si \rangle & \langle -si, \text{molu-} \rangle & \langle -si, -ess \rangle & \langle -si, -ta \rangle & & \\ \langle \text{molu-}, \text{molu-} \rangle & \langle \text{molu-}, \emptyset \rangle & \langle \text{molu-}, -si \rangle & \langle \text{molu-}, \text{molu-} \rangle & \langle \text{molu-}, -ess \rangle & \langle \text{molu-}, -ta \rangle & & \\ \langle -ess, \text{molu-} \rangle & \langle -ess, \emptyset \rangle & \langle -ess, -si \rangle & \langle -ess, \text{molu-} \rangle & \langle -ess, -ess \rangle & \langle -ess, -ta \rangle & & \end{array} \right\}$$

Table 17: The unrepaired, unsimplified s-precedence relation for (24). To save space, DP_s and DP_o are written in place of *halapeci-kkeyse* and *Inbo-lul*, respectively.

$$\left\{ \begin{array}{cccccc} \langle \text{DP}_s, \text{DP}_s \rangle & \langle \text{DP}_s, \text{DP}_o \rangle & \langle \text{DP}_s, \text{molu-} \rangle & \langle \text{DP}_s, \emptyset \rangle & \langle \text{DP}_s, -si \rangle & \langle \text{DP}_s, -ess \rangle & \langle \text{DP}_s, -ta \rangle \\ \langle \text{DP}_o, \text{molu-} \rangle & \langle \text{DP}_o, \emptyset \rangle & \langle \text{DP}_o, -si \rangle & \langle \text{DP}_o, -ess \rangle & \langle \text{DP}_o, -ta \rangle & & \\ \langle \text{molu-}, \text{molu-} \rangle & \langle \text{molu-}, \emptyset \rangle & \langle \text{molu-}, -si \rangle & \langle \text{molu-}, -ess \rangle & \langle \text{molu-}, -ta \rangle & & \\ \langle \emptyset, \text{molu-} \rangle & \langle \emptyset, \emptyset \rangle & \langle \emptyset, -si \rangle & \langle \emptyset, -ess \rangle & \langle \emptyset, -ta \rangle & & \\ \langle -si, \text{molu-} \rangle & \langle -si, \emptyset \rangle & \langle -si, -si \rangle & \langle -si, -ess \rangle & \langle -si, -ta \rangle & & \\ \langle -ess, \text{molu-} \rangle & \langle -ess, \emptyset \rangle & \langle -ess, -si \rangle & \langle -ess, -ess \rangle & \langle -ess, -ta \rangle & & \end{array} \right\}$$

Table 18: The unrepaired, simplified s-precedence relation for (24). To save space, DP_s and DP_o are written in place of *halapeci-kkeyse* and *Inbo-lul*, respectively.

$$\left\{ \begin{array}{ccccccc} \langle \text{DP}_s, \text{DP}_s \rangle & \langle \text{DP}_s, \text{DP}_o \rangle & \langle \text{DP}_s, \textit{molu-} \rangle & \langle \text{DP}_s, \emptyset \rangle & \langle \text{DP}_s, \textit{-si} \rangle & \langle \text{DP}_s, \textit{-ess} \rangle & \langle \text{DP}_s, \textit{-ta} \rangle \\ & & \langle \text{DP}_o, \textit{molu-} \rangle & \langle \text{DP}_o, \emptyset \rangle & \langle \text{DP}_o, \textit{-si} \rangle & \langle \text{DP}_o, \textit{-ess} \rangle & \langle \text{DP}_o, \textit{-ta} \rangle \\ & & \langle \textit{molu-}, \textit{molu-} \rangle & \langle \textit{molu-}, \emptyset \rangle & \langle \textit{molu-}, \textit{-si} \rangle & \langle \textit{molu-}, \textit{-ess} \rangle & \langle \textit{molu-}, \textit{-ta} \rangle \\ & & & \langle \emptyset, \emptyset \rangle & \langle \emptyset, \textit{-si} \rangle & \langle \emptyset, \textit{-ess} \rangle & \langle \emptyset, \textit{-ta} \rangle \\ & & & & \langle \textit{-si}, \textit{-si} \rangle & \langle \textit{-si}, \textit{-ess} \rangle & \langle \textit{-si}, \textit{-ta} \rangle \\ & & & & & \langle \textit{-ess}, \textit{-ess} \rangle & \langle \textit{-ess}, \textit{-ta} \rangle \end{array} \right\}$$

Table 19: The repaired s-precedence relation for (24). To save space, DP_s and DP_o are written in place of *halapeci-kkeyse* and *Inholul*, respectively.

This example shows that suppletion is possible even in the absence of adjacency. To demonstrate that the symmetry produced by displacement is required for this sort of suppletion (or portmanteau), we will consider the case of Korean “long-form” (periphrastic) negation. Negative suppletion is not possible in this construction (Choi & Harley, 2019):

- (26) a. Inho-nun Mina-lul al-ci mos hay-ess-ta
 Inho-TOP Mina-ACC know-CI NEG do-PST-DECL
 ‘Inho didn’t know Mina’ (Choi & Harley, 2019, (47c))
- b. *Inho-nun Mina-lul molu-ci hay-ess-ta
 Inho-TOP Mina-ACC know.NEG-CI do-PST-DECL

We assume for (26) the structure in Figure 7, with the node realized as *-ci* occupying a position between Hon and Neg, for reasons to be made clear in section 3.5.²⁰ Under this analysis, long-form negation involves cyclic head-movement of *v* to C while the verbal root remains *in situ*. This differs from Choi & Harley’s (2019) analysis, in which Neg moves to C and a *v* is adjoined to Neg post-syntactically. However, that analysis is incompatible with our framework, which does not adopt node-sprouting (post-syntactic insertion of nodes).

When the structure in Figure 7 is linearized, it will produce the u-precedence relation in Table 20. Assuming that Vocabulary Insertion provides the exponents in (27), the u-precedence relation is mapped onto the s-precedence relation in Table 21, which simplifies to Table 22 after removing repetitions.

(27) Vocabulary mapping for (26).

- a. $DP_s \rightarrow \text{Inho-nun}$
- b. $DP_o \rightarrow \text{Mina-lul}$
- c. $\sqrt{\text{KNOW}} \rightarrow \text{molu}_1$
- d. $v \rightarrow \text{hay-}$
- e. $\text{Hon} \rightarrow \emptyset$

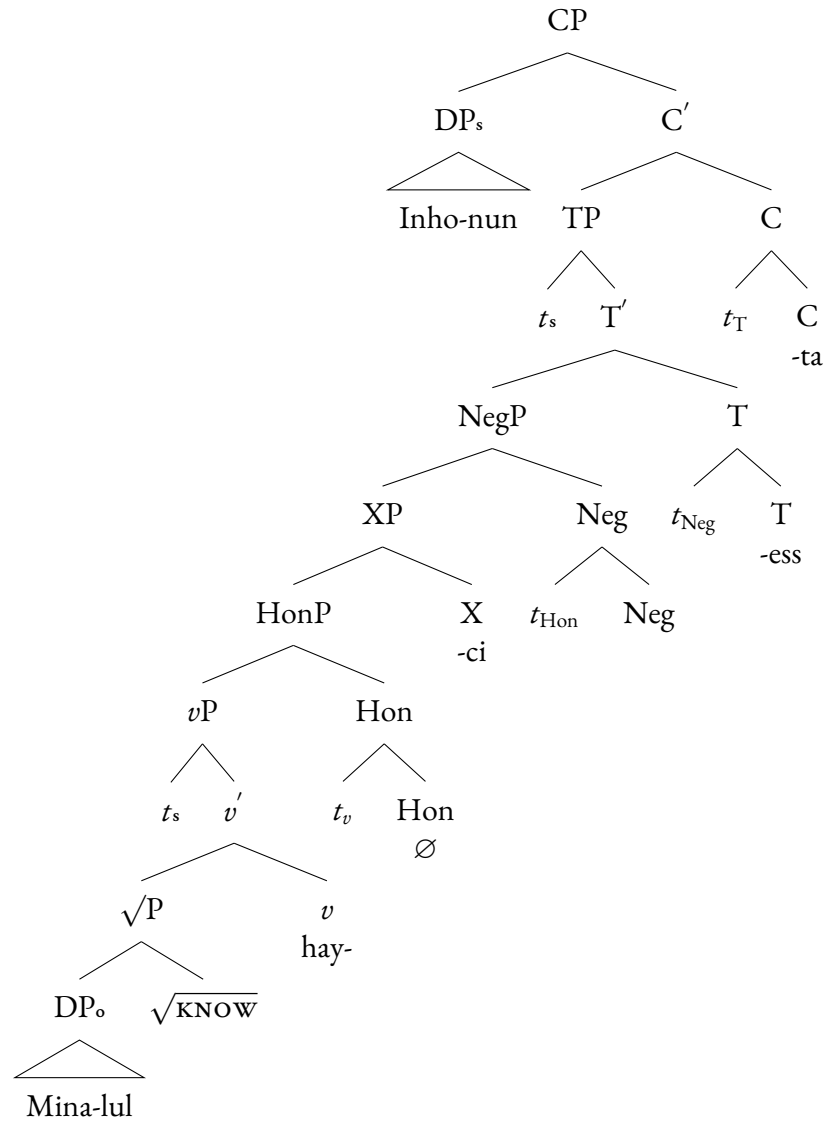


Figure 7: The syntactic structure assumed for (26). For clarity, Vocabulary Items are included for nodes not involved in suppletion. Traces are used as a notational device to indicate movement.

- f. $X \rightarrow -ci$
- g. $Neg \rightarrow molu_{-1}$
- h. $T \rightarrow -ess$
- i. $C \rightarrow -ta$

This s-precedence relation is not antisymmetric. Of particular import, it contains both $\langle molu-, -ci \rangle$ and $\langle -ci, molu- \rangle$. The former pair realizes $\langle \sqrt{KNOW}, X \rangle$, which has no counterpart $^*\langle X, \sqrt{KNOW} \rangle$. By the Principle of Maximality, then, $\langle \sqrt{KNOW}, X \rangle$ cannot be deleted. Conversely, $\langle -ci, molu- \rangle$ realizes $\langle X, Neg \rangle$, which has no counterpart $^*\langle Neg, X \rangle$. Consequently, $\langle -ci, molu- \rangle$ cannot be deleted either. Therefore, there is no way to repair this s-precedence relation, and (26b) is correctly predicted to be ill-formed.

{	$\langle \text{DP}_s, \text{DP}_s \rangle$	$\langle \text{DP}_s, \text{DP}_o \rangle$	$\langle \text{DP}_s, \sqrt{\text{KNOW}} \rangle$	$\langle \text{DP}_s, v \rangle$	$\langle \text{DP}_s, \text{Hon} \rangle$	$\langle \text{DP}_s, X \rangle$	$\langle \text{DP}_s, \text{Neg} \rangle$	$\langle \text{DP}_s, T \rangle$	$\langle \text{DP}_s, C \rangle$
			$\langle \text{DP}_o, \sqrt{\text{KNOW}} \rangle$	$\langle \text{DP}_o, v \rangle$	$\langle \text{DP}_o, \text{Hon} \rangle$	$\langle \text{DP}_o, X \rangle$	$\langle \text{DP}_o, \text{Neg} \rangle$	$\langle \text{DP}_o, T \rangle$	$\langle \text{DP}_o, C \rangle$
				$\langle \sqrt{\text{KNOW}}, v \rangle$	$\langle \sqrt{\text{KNOW}}, \text{Hon} \rangle$	$\langle \sqrt{\text{KNOW}}, X \rangle$	$\langle \sqrt{\text{KNOW}}, \text{Neg} \rangle$	$\langle \sqrt{\text{KNOW}}, T \rangle$	$\langle \sqrt{\text{KNOW}}, C \rangle$
				$\langle v, v \rangle$	$\langle v, \text{Hon} \rangle$	$\langle v, X \rangle$	$\langle v, \text{Neg} \rangle$	$\langle v, T \rangle$	$\langle v, C \rangle$
				$\langle \text{Hon}, v \rangle$	$\langle \text{Hon}, \text{Hon} \rangle$	$\langle \text{Hon}, X \rangle$	$\langle \text{Hon}, \text{Neg} \rangle$	$\langle \text{Hon}, T \rangle$	$\langle \text{Hon}, C \rangle$
				$\langle X, v \rangle$	$\langle X, \text{Hon} \rangle$		$\langle X, \text{Neg} \rangle$	$\langle X, T \rangle$	$\langle X, C \rangle$
				$\langle \text{Neg}, v \rangle$	$\langle \text{Neg}, \text{Hon} \rangle$		$\langle \text{Neg}, \text{Neg} \rangle$	$\langle \text{Neg}, T \rangle$	$\langle \text{Neg}, C \rangle$
				$\langle T, v \rangle$	$\langle T, \text{Hon} \rangle$		$\langle T, \text{Neg} \rangle$	$\langle T, T \rangle$	$\langle T, C \rangle$

Table 20: The u-precedence relation for the structure in Figure 7.

{	$\langle DP_s, DP_s \rangle$	$\langle DP_s, DP_o \rangle$	$\langle DP_s, molu- \rangle$	$\langle DP_s, hay- \rangle$	$\langle DP_s, \emptyset \rangle$	$\langle DP_s, -ci \rangle$	$\langle DP_s, molu- \rangle$	$\langle DP_s, -ess \rangle$	$\langle DP_s, -ta \rangle$
			$\langle DP_o, molu- \rangle$	$\langle DP_o, hay- \rangle$	$\langle DP_o, \emptyset \rangle$	$\langle DP_o, -ci \rangle$	$\langle DP_o, molu- \rangle$	$\langle DP_o, -ess \rangle$	$\langle DP_o, -ta \rangle$
				$\langle molu-, hay- \rangle$	$\langle molu-, \emptyset \rangle$	$\langle molu-, -ci \rangle$	$\langle molu-, molu- \rangle$	$\langle molu-, -ess \rangle$	$\langle molu-, -ta \rangle$
				$\langle hay-, hay- \rangle$	$\langle hay-, \emptyset \rangle$	$\langle hay-, -ci \rangle$	$\langle hay-, molu- \rangle$	$\langle hay-, -ess \rangle$	$\langle hay-, -ta \rangle$
				$\langle \emptyset, hay- \rangle$	$\langle \emptyset, \emptyset \rangle$	$\langle \emptyset, -ci \rangle$	$\langle \emptyset, molu- \rangle$	$\langle \emptyset, -ess \rangle$	$\langle \emptyset, -ta \rangle$
				$\langle -ci, hay- \rangle$	$\langle -ci, \emptyset \rangle$		$\langle -ci, molu- \rangle$	$\langle -ci, -ess \rangle$	$\langle -ci, -ta \rangle$
				$\langle molu-, hay- \rangle$	$\langle molu-, \emptyset \rangle$		$\langle molu-, molu- \rangle$	$\langle molu-, -ess \rangle$	$\langle molu-, -ta \rangle$
				$\langle -ess, hay- \rangle$	$\langle -ess, \emptyset \rangle$		$\langle -ess, molu- \rangle$	$\langle -ess, -ess \rangle$	$\langle -ess, -ta \rangle$

Table 21: The unsimplified s-precedence relation for (26b). To save space, DP_s and DP_o are written in place of *Inho-nun* and *Minalul*, respectively.

{	$\langle DP_s, DP_s \rangle$	$\langle DP_s, DP_o \rangle$	$\langle DP_s, molu- \rangle$	$\langle DP_s, hay- \rangle$	$\langle DP_s, \emptyset \rangle$	$\langle DP_s, -ci \rangle$	$\langle DP_s, molu- \rangle$	$\langle DP_s, -ess \rangle$	$\langle DP_s, -ta \rangle$
			$\langle DP_o, molu- \rangle$	$\langle DP_o, hay- \rangle$	$\langle DP_o, \emptyset \rangle$	$\langle DP_o, -ci \rangle$	$\langle DP_o, molu- \rangle$	$\langle DP_o, -ess \rangle$	$\langle DP_o, -ta \rangle$
				$\langle molu-, hay- \rangle$	$\langle molu-, \emptyset \rangle$	$\langle molu-, -ci \rangle$	$\langle molu-, molu- \rangle$	$\langle molu-, -ess \rangle$	$\langle molu-, -ta \rangle$
				$\langle hay-, hay- \rangle$	$\langle hay-, \emptyset \rangle$	$\langle hay-, -ci \rangle$	$\langle hay-, molu- \rangle$	$\langle hay-, -ess \rangle$	$\langle hay-, -ta \rangle$
				$\langle \emptyset, hay- \rangle$	$\langle \emptyset, \emptyset \rangle$	$\langle \emptyset, -ci \rangle$	$\langle \emptyset, molu- \rangle$	$\langle \emptyset, -ess \rangle$	$\langle \emptyset, -ta \rangle$
				$\langle -ci, hay- \rangle$	$\langle -ci, \emptyset \rangle$		$\langle -ci, molu- \rangle$	$\langle -ci, -ess \rangle$	$\langle -ci, -ta \rangle$
				$\langle -ess, hay- \rangle$	$\langle -ess, \emptyset \rangle$		$\langle -ess, molu- \rangle$	$\langle -ess, -ess \rangle$	$\langle -ess, -ta \rangle$

Table 22: The simplified s-precedence relation for (26b). To save space, DP_s and DP_o are written in place of *Inho-nun* and *Minalul*, respectively.

In both (24) and (26), $\sqrt{\text{KNOW}}$ and Neg are not adjacent. However, in the former case, negative suppletion is still possible, while in the latter case it is not. The crucial difference between these examples is that in the former, $\sqrt{\text{KNOW}}$ moves above the base-merge position of Neg, while in the latter, it remains *in situ* (with the presence of *-ci* further blocking an adjacency-based portmanteau).²¹ This demonstrates that the set of ordered pairs produced by crossover indeed allows for the insertion of portmanteau morphemes.

3.4.3 Summary

To summarize, our proposal predicts that portmanteau morphemes (including suppletive allomorphs) can be inserted in two structural configurations.^{22, 23} First, a portmanteau can be inserted for linearly adjacent nodes. Second, a portmanteau can be inserted when one node has moved past another, even if they would not otherwise be expected to be adjacent. Importantly, this account of portmanteaux requires no additional assumptions beyond the two theses of this article: that linear order is antisymmetric, and that Vocabulary Insertion allows many-to-many mappings. In a sense, the terminal node remains the unit of Vocabulary Insertion, but Vocabulary Insertion can insert the same Vocabulary Item for different terminal nodes, so that terminal nodes share an exponent.²⁴ Crucially, this is not an *additional* assumption about Vocabulary Insertion, but simply a *different* one from that made by frameworks that assume Vocabulary Insertion to be one-to-one (see section 2.2). Additionally, we are not assuming that sequences of linearly adjacent nodes (or sets of nodes in symmetric precedence relationships) are the targets of Vocabulary Insertion. In other words, we are not proposing a novel primitive as the target for Vocabulary Insertion comparable to the span (Svenonius, 2016) or stretch (Ostrove, 2018). In principle, Vocabulary Insertion can insert a portmanteau for any set of nodes, regardless of the structural relations between them. However, insertion of portmanteaux is constrained by outside factors. First, Vocabulary Insertion can naturally only insert a portmanteau if one is available in the lexicon. Second, because of the constraints imposed by linearization—

the s-precedence relation must be transitive, total and antisymmetric, and it must reflect the u-precedence relation (as expressed by the Principle of Maximality)—it follows that portmanteaux can only be inserted for linearly adjacent nodes and for nodes in a crossover relationship, as demonstrated above. In general, trying to insert portmanteaux in other configurations will cause linearization to fail.

Because it depends only on Vocabulary Insertion and linearization, both independently motivated²⁵ features of the linguistic system, the present proposal is conceptually preferable to other proposals that posit morphology-specific units or processes to derive portmanteaux. As already discussed, this proposal does not posit a morphology-specific unit of Vocabulary Insertion comparable to the span or stretch. Unlike in Distributed Morphology, we do not need to posit a morphological operation of Fusion (Halle & Marantz, 1993). And unlike versions of Nanosyntax that adopt non-terminal spell-out, we do not need to posit otherwise unmotivated traceless movement (“spell-out-driven movement”) to derive the right structural configuration for Vocabulary Insertion (Baunaz & Lander, 2018). We rely only on the independently motivated operations of linearization and Vocabulary Insertion.

3.5 Many-to-many mappings

In previous sections, we have demonstrated how our proposal accounts for one-to-many mappings (i.e., multiple exponence; section 3.2) and many-to-one mappings (i.e., portmanteaux and suppletion; section 3.4). In this final section, we demonstrate many-to-many mappings, where both occur, using an additional example from Korean. In addition to negative suppletion, Korean also has verbs with suppletive forms used in honorific contexts (Chung, 2009). Unlike negative suppletion, honorific suppletion is possible in long-form negation, and co-occurs with separative honorific marking on the auxiliary (Choi & Harley, 2019):

- (28) Halapeci-kkeyse cwumwusi-ci an(i) ha-si-ess-ta
 grandfather-NOM.HON sleep.HON-CI NEG do-HON-PST-DECL

‘Grandfather didn’t sleep’ (Choi & Harley, 2019, (46a))

This is a many-to-many mapping (specifically, partially superfluous multiple exponence; Caballero & Harris, 2012) with a feature (Hon) corresponding to multiple lexical items and a lexical item (*cwumwusi*) corresponding to multiple features.

We assume for (28) the structure in Figure 8, with movement the subject from the Spec,vP to Spec,CP via Spec,TP and head movement of $\sqrt{\text{SLEEP}}$ to C via the intermediate heads, but skipping X, as in Figure 7 (see footnote 20). Linearizing this structure will produce the u-precedence relation in Table 23.

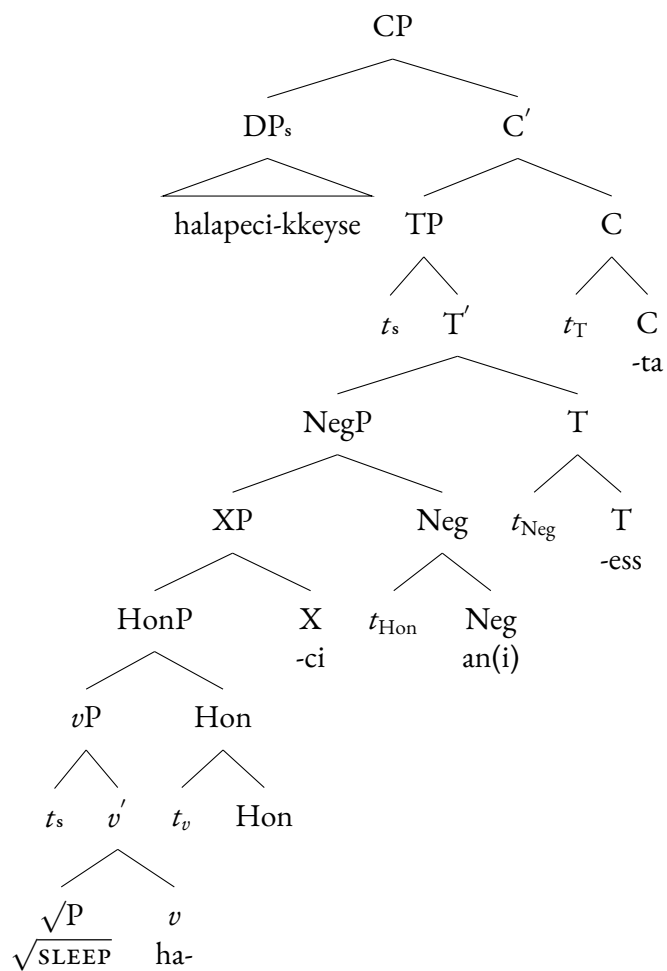


Figure 8: The syntactic structure assumed for (28). For clarity, Vocabulary Items are included for nodes not involved in suppletion. Traces are used as a notational device to indicate movement.

$\langle \text{DP}_s, \text{DP}_s \rangle$	$\langle \text{DP}_s, \sqrt{\text{SLEEP}} \rangle$	$\langle \text{DP}_s, v \rangle$	$\langle \text{DP}_s, \text{Hon} \rangle$	$\langle \text{DP}_s, \text{X} \rangle$	$\langle \text{DP}_s, \text{Neg} \rangle$	$\langle \text{DP}_s, \text{T} \rangle$	$\langle \text{DP}_s, \text{C} \rangle$
		$\langle \sqrt{\text{SLEEP}}, v \rangle$	$\langle \sqrt{\text{SLEEP}}, \text{Hon} \rangle$	$\langle \sqrt{\text{SLEEP}}, \text{X} \rangle$	$\langle \sqrt{\text{SLEEP}}, \text{Neg} \rangle$	$\langle \sqrt{\text{SLEEP}}, \text{T} \rangle$	$\langle \sqrt{\text{SLEEP}}, \text{C} \rangle$
		$\langle v, v \rangle$	$\langle v, \text{Hon} \rangle$	$\langle v, \text{X} \rangle$	$\langle v, \text{Neg} \rangle$	$\langle v, \text{T} \rangle$	$\langle v, \text{C} \rangle$
		$\langle \text{Hon}, v \rangle$	$\langle \text{Hon}, \text{Hon} \rangle$	$\langle \text{Hon}, \text{X} \rangle$	$\langle \text{Hon}, \text{Neg} \rangle$	$\langle \text{Hon}, \text{T} \rangle$	$\langle \text{Hon}, \text{C} \rangle$
		$\langle \text{X}, v \rangle$	$\langle \text{X}, \text{Hon} \rangle$		$\langle \text{X}, \text{Neg} \rangle$	$\langle \text{X}, \text{T} \rangle$	$\langle \text{X}, \text{C} \rangle$
		$\langle \text{Neg}, v \rangle$	$\langle \text{Neg}, \text{Hon} \rangle$		$\langle \text{Neg}, \text{Neg} \rangle$	$\langle \text{Neg}, \text{T} \rangle$	$\langle \text{Neg}, \text{C} \rangle$
		$\langle \text{T}, v \rangle$	$\langle \text{T}, \text{Hon} \rangle$		$\langle \text{T}, \text{Neg} \rangle$	$\langle \text{T}, \text{T} \rangle$	$\langle \text{T}, \text{C} \rangle$

Table 23: The u-precedence relation for the structure in Figure 8.

If Vocabulary Insertion inserts the lexical items in (29), then the u-precedence relation in Table 23 will be mapped onto the s-precedence relation in Table 24, which simplifies to Table 25 after removing repetitions.

(29) Vocabulary mapping for (28).

- a. $DP_s \rightarrow \textit{halapeci-kkeyse}$
- b. $\sqrt{\text{SLEEP}} \rightarrow \textit{cwumwusi-}_1$
- c. $v \rightarrow \textit{ha-}$
- d. $\text{Hon} \rightarrow \textit{cwumwusi-}_1$
- e. $\text{Hon} \rightarrow \textit{-si}$
- f. $X \rightarrow \textit{-ci}$
- g. $\text{Neg} \rightarrow \textit{an(i)}$
- h. $T \rightarrow \textit{-ess}$
- i. $C \rightarrow \textit{-ta}$

The s-precedence relation is not asymmetric, but it is able to be repaired. Notably, unlike in the case of negative suppletion discussed above, there is no ordering paradox between the verb and *-ci*. Although the s-precedence relation contains both $\langle \textit{cwumwusi-}, \textit{-ci} \rangle$ and $\langle \textit{-ci}, \textit{cwumwusi-} \rangle$, the latter realizes $\langle X, \text{Hon} \rangle$. This pair does have a symmetric counterpart $\langle \text{Hon}, X \rangle$, so the Principle of Maximality does not protect $\langle \textit{-ci}, \textit{cwumwusi-} \rangle$. Consequently, this pair can be deleted, allowing *cwumwusi-* to be realized before *-ci*. Deleting the underlined ordered pairs leaves the s-precedence relation in Table 26, as desired.²⁶ This demonstrates that the present proposal indeed allows many-to-many mappings of syntactic structure to lexical items.

$$\left\{ \begin{array}{cccccccccc} \langle \text{DP}_s, \text{DP}_s \rangle & \langle \text{DP}_s, \text{V}_{\text{Hon}} \rangle & \langle \text{DP}_s, ha- \rangle & \langle \text{DP}_s, \text{V}_{\text{Hon}} \rangle & \langle \text{DP}_s, -si \rangle & \langle \text{DP}_s, -ci \rangle & \langle \text{DP}_s, an(i) \rangle & \langle \text{DP}_s, -ess \rangle & \langle \text{DP}_s, -ta \rangle \\ \langle \text{V}_{\text{Hon}}, ha- \rangle & \langle \text{V}_{\text{Hon}}, \text{V}_{\text{Hon}} \rangle & \langle \text{V}_{\text{Hon}}, -si \rangle & \langle \text{V}_{\text{Hon}}, -ci \rangle & \langle \text{V}_{\text{Hon}}, an(i) \rangle & \langle \text{V}_{\text{Hon}}, -ess \rangle & \langle \text{V}_{\text{Hon}}, -ta \rangle \\ \langle ha-, ha- \rangle & \langle ha-, \text{V}_{\text{Hon}} \rangle & \langle ha-, -si \rangle & \langle ha-, -ci \rangle & \langle ha-, an(i) \rangle & \langle ha-, -ess \rangle & \langle ha-, -ta \rangle \\ \langle \text{V}_{\text{Hon}}, ha- \rangle & \langle \text{V}_{\text{Hon}}, \text{V}_{\text{Hon}} \rangle & \langle \text{V}_{\text{Hon}}, -si \rangle & \langle \text{V}_{\text{Hon}}, -ci \rangle & \langle \text{V}_{\text{Hon}}, an(i) \rangle & \langle \text{V}_{\text{Hon}}, -ess \rangle & \langle \text{V}_{\text{Hon}}, -ta \rangle \\ \langle -si, ha- \rangle & \langle -si, \text{V}_{\text{Hon}} \rangle & \langle -si, -si \rangle & \langle -si, -ci \rangle & \langle -si, an(i) \rangle & \langle -si, -ess \rangle & \langle -si, -ta \rangle \\ \langle -ci, ha- \rangle & \langle -ci, \text{V}_{\text{Hon}} \rangle & \langle -ci, -si \rangle & & \langle -ci, an(i) \rangle & \langle -ci, -ess \rangle & \langle -ci, -ta \rangle \\ \langle an(i), ha- \rangle & \langle an(i), \text{V}_{\text{Hon}} \rangle & \langle an(i), -si \rangle & & \langle an(i), an(i) \rangle & \langle an(i), -ess \rangle & \langle an(i), -ta \rangle \\ \langle -ess, ha- \rangle & \langle -ess, \text{V}_{\text{Hon}} \rangle & \langle -ess, -si \rangle & & \langle -ess, an(i) \rangle & \langle -ess, -ess \rangle & \langle -ess, -ta \rangle \end{array} \right\}$$

Table 24: The unsimplified, unrepaired s-precedence relation for (28). To save space, DP_s and V_{Hon} are written in place of *halapeci-keyse* and *cwumwusi-*, respectively.

$$\left\{ \begin{array}{cccccccc} \langle \text{DP}_s, \text{DP}_s \rangle & \langle \text{DP}_s, ha- \rangle & \langle \text{DP}_s, \text{V}_{\text{Hon}} \rangle & \langle \text{DP}_s, -si \rangle & \langle \text{DP}_s, -ci \rangle & \langle \text{DP}_s, an(i) \rangle & \langle \text{DP}_s, -ess \rangle & \langle \text{DP}_s, -ta \rangle \\ \langle \text{V}_{\text{Hon}}, ha- \rangle & \langle \text{V}_{\text{Hon}}, \text{V}_{\text{Hon}} \rangle & \langle \text{V}_{\text{Hon}}, -si \rangle & \langle \text{V}_{\text{Hon}}, -ci \rangle & \langle \text{V}_{\text{Hon}}, an(i) \rangle & \langle \text{V}_{\text{Hon}}, -ess \rangle & \langle \text{V}_{\text{Hon}}, -ta \rangle \\ \langle ha-, ha- \rangle & \langle ha-, \text{V}_{\text{Hon}} \rangle & \langle ha-, -si \rangle & \langle ha-, -ci \rangle & \langle ha-, an(i) \rangle & \langle ha-, -ess \rangle & \langle ha-, -ta \rangle \\ \langle -si, ha- \rangle & \langle -si, \text{V}_{\text{Hon}} \rangle & \langle -si, -si \rangle & \langle -si, -ci \rangle & \langle -si, an(i) \rangle & \langle -si, -ess \rangle & \langle -si, -ta \rangle \\ \langle -ci, ha- \rangle & \langle -ci, \text{V}_{\text{Hon}} \rangle & \langle -ci, -si \rangle & & \langle -ci, an(i) \rangle & \langle -ci, -ess \rangle & \langle -ci, -ta \rangle \\ \langle an(i), ha- \rangle & \langle an(i), \text{V}_{\text{Hon}} \rangle & \langle an(i), -si \rangle & & \langle an(i), an(i) \rangle & \langle an(i), -ess \rangle & \langle an(i), -ta \rangle \\ \langle -ess, ha- \rangle & \langle -ess, \text{V}_{\text{Hon}} \rangle & \langle -ess, -si \rangle & & \langle -ess, an(i) \rangle & \langle -ess, -ess \rangle & \langle -ess, -ta \rangle \end{array} \right\}$$

Table 25: The simplified, unrepaired s-precedence relation for (28). To save space, DP_s and V_{Hon} are written in place of *halapeci-keyse* and *cwumwusi-*, respectively. Underlined ordered pairs will be deleted to repair the relation.

$$\left\{ \begin{array}{cccccccc} \langle \text{DP}_s, \text{DP}_s \rangle & \langle \text{DP}_s, \text{V}_{\text{Hon}} \rangle & \langle \text{DP}_s, -ci \rangle & \langle \text{DP}_s, an(i) \rangle & \langle \text{DP}_s, ha- \rangle & \langle \text{DP}_s, -si \rangle & \langle \text{DP}_s, -ess \rangle & \langle \text{DP}_s, -ta \rangle \\ & \langle \text{V}_{\text{Hon}}, \text{V}_{\text{Hon}} \rangle & \langle \text{V}_{\text{Hon}}, -ci \rangle & \langle \text{V}_{\text{Hon}}, an(i) \rangle & \langle \text{V}_{\text{Hon}}, ha- \rangle & \langle \text{V}_{\text{Hon}}, -si \rangle & \langle \text{V}_{\text{Hon}}, -ess \rangle & \langle \text{V}_{\text{Hon}}, -ta \rangle \\ & & & \langle -ci, an(i) \rangle & \langle -ci, ha- \rangle & \langle -ci, -si \rangle & \langle -ci, -ess \rangle & \langle -ci, -ta \rangle \\ & & & \langle an(i), an(i) \rangle & \langle an(i), ha- \rangle & \langle an(i), -si \rangle & \langle an(i), -ess \rangle & \langle an(i), -ta \rangle \\ & & & & \langle ha-, ha- \rangle & \langle ha-, -si \rangle & \langle ha-, -ess \rangle & \langle ha-, -ta \rangle \\ & & & & & \langle -si, -si \rangle & \langle -si, -ess \rangle & \langle -si, -ta \rangle \\ & & & & & & \langle -ess, -ess \rangle & \langle -ess, -ta \rangle \end{array} \right\}$$

Table 26: The repaired s-precedence relation for (28). To save space, DP_s and V_{Hon} are written in place of *halapeci-kkeyse* and *cwumwusi-*, respectively. The rows and columns have been reordered to make the ordering clearer.

4 Discussion

In this article, we have argued for two proposals in service of advancing a more restrictive theory of externalization. The first proposal is that linear order is better modeled as antisymmetric than asymmetric. Conceptually, this proposal reflects changes to the theoretical landscape since Kayne 1994. In effect, a side effect of the reframing of movement as Merge is that reflexivity becomes endemic to syntax. If we assume that linear order is asymmetric, then such reflexivity poses a problem when syntactic structure is linearized. However, the problem can be avoided if linear order is taken to be antisymmetric, thereby allowing reflexivity. The main empirical advantage of antisymmetry over asymmetry is its ability to account for simultaneity. Here, we discussed grammatical tone, non-manual signs, and portmanteaux, but the proposal can be extended to other simultaneity phenomena. For example, it can potentially be extended to infixation and reduplication, allowing these phenomena to be generated as part of linearization, rather than requiring separate processes.²⁷ For example, if an s-precedence relation contains $\langle infix, base \rangle$, then the Precedence Mapping Principle may allow an infxal realization as $b\langle infix \rangle ase$, depending on how we define precedence. Specifically at issue is whether, if *infix* precedes *base*, every segment or feature of *infix* must precede every segment or feature of *base*. If not, then infixation is possible. Additionally, this proposal should be compatible with approaches that treat (components of) intonational contours as exponents of syntactic structure. This is no small advantage: if we count intonation, then simultaneity is a universal or near-universal property of language. Conversely, there appear to be no advantages of adopting asymmetry over antisymmetry. One might argue that antisymmetry introduces a complication in that linearization needs to determine whether to create a sequential or simultaneous representation. However, if simultaneity simply occurs wherever it is possible (Kremers, 2013)—for example, where material occupies different autosegmental tiers—then there is no burden of deciding between simultaneity and sequentiality.

The second proposal is that Vocabulary Insertion is not one-to-one, but many-to-many. Despite the well-known fact that the relationship between syntactic features and their exponents is many-to-many, a number of morphological theories assume the mechanism behind this relationship (i.e., Vocabulary Insertion) to be one-to-one. Such theories must then propose additional mechanisms to account for deviations from this idealized one-to-one relationship. By contrast, we attribute the many-to-many relationship between syntactic features and exponents to Vocabulary Insertion itself, obviating the need for additional mechanisms. Consequently, we can generate portmanteaux and multiple exponence without adopting (i) morphology-specific operations such as Fission or Fusion, or (ii) morphology-specific units such as the span or stretch.

Allowing Vocabulary Insertion to be many-to-many makes the operation more powerful, potentially raising concerns about overgeneration. However, we have demonstrated that linearization constrains Vocabulary Insertion: under the wrong conditions, insertion of portmanteaux or multiple exponents causes linearization to fail. We also suggested other factors that may constrain Vocabulary Insertion, although these were not the focus of the article. For example, we suggested that parsing and economy may favor single exponence. On the other hand, multiple exponence may be favored because it arguably benefits the listener: it provides more opportunities to perceive content, and in some cases makes the syntactic representation more explicit (when “traces” are pronounced, as in (14)). This picture of competing constraints influencing Vocabulary Insertion suggests an Optimality Theoretic approach.²⁸ This would bring Vocabulary Insertion in line with other postsyntactic phenomena: Optimality Theory is mainstream in treatments of phonology proper and of prosodic parsing (Selkirk, 2011), and is gaining ground in studies of linearization (Kusmer, 2020; Johnson, 2023).

The proposals in this article constitute a first step towards a novel research program, which we call *Intermodular Morphology*. The goal of this research program is to capture morphological generalizations without a morphology module. In effect, there is syntax, there is phonology,

and there are the intermodular operations that translate between them—in other words, the syntax-phonology interface or “PF.” Specifically, Intermodular Morphology rejects PF operations that are not *directly implicated* in the translation from syntax to phonology. Thus, PF operations must have the following properties:

(30) Properties of PF operations in Intermodular Morphology

- a. The operation takes as its input (i) the output of syntax or (ii) the output of an earlier PF operation.²⁹
- b. The output of the operation is not a syntactic representation.

Intermodular Morphology thus excludes most postsyntactic operations of Distributed Morphology, as well the spell-out driven movement of Nanosyntax, since these are non-syntactic operations that produce syntactic representations as outputs, in violation of (30b). Likewise, because there is no morphology component, there are no representations proper to that component. Consequently, there can be no spans unless they are produced as the output of an intermodular operation. Intermodular Morphology thus avoids the conceptual challenges faced by these theoretical tools (see section 2.2 and Collins & Kayne, 2023) and presents a more parsimonious view of Language, with fewer modules and fewer primitives. Moreover, by reducing the number of unexplained primitives, Intermodular Morphology forces us to go beyond explanatory adequacy (Chomsky, 2004) in our account of morphological phenomena.

In this article, we have taken linearization and Vocabulary Insertion to be intermodular, licensing their use in accounting for morphological generalizations. Linearization takes the syntactic structure as its input, satisfying (30a), and produces the u-precedence relation as its output. The u-precedence relation is a set of ordered pairs of terminal nodes, which is not a syntactic representation, thereby satisfying (30b).³⁰ Likewise, Vocabulary Insertion takes as its inputs the u-precedence relation and the syntactic structure (since features are used in selecting exponents) and produces the s-precedence relation. The s-precedence relation is a set of ordered

pairs of exponents, which is not a syntactic object.

Intermodular Morphology is a research program, not a specific framework. Thus, the proposals advanced in this article represent only one implementation of Intermodular Morphology. For example, other researchers may assume a different inventory of intermodular operations, or may assume these operations to be of a different nature. Additionally, researchers may vary in whether they attribute a specific phenomenon to the syntax, the phonology, intermodular operations, or some combination of these. For example, Collins & Kayne's (2023) *Morphology-as-Syntax* rejects postsyntactic Vocabulary Insertion, and in general, it seeks morphological explanations in the syntax. Despite these differences, *Morphology-as-Syntax* can be considered an implementation of Intermodular Morphology: it accounts for morphological generalizations without a morphology module. We hope that readers will take seriously the pursuit of an Intermodular Morphology, even if they do not agree with the specific proposals advanced in this article.

References

- Arregi, Karlos & Andrew Nevins. 2012. *Morphotactics: Basque auxiliaries and the structure of Spellout*. Dordrecht: Springer.
- Baunaz, Lena & Eric Lander. 2018. Nanosyntax: The basics. In Lena Baunaz, Liliane Haegeman, Karen De Clercq & Eric Lander (eds.), *Exploring nanosyntax*, 3–56. Oxford: Oxford University Press.
- Caballero, Gabriela & Alice C. Harris. 2012. A working typology of multiple exponence. In Ferenc Kiefer, Mária Ladányi & Péter Siptár (eds.), *Current issues in morphological theory: (ir)regularity, analogy and frequency*, 163–188. Amsterdam: John Benjamins.
- Caha, Pavel. 2009. *The nanosyntax of case*. Tromsø: University of Tromsø PhD thesis.
- Carstens, Vicki. 2011. Hyperactivity and hyperagreement in Bantu. *Lingua* 121. 721–741.
- Choi, Jaehoon & Heidi Harley. 2019. Locality domains and morphological rules: Phases, heads, node-sprouting and suppletion in Korean honorification. *Natural Language & Linguistic Theory* 37. 1319–1365.
- Chomsky, Noam. 1995. Bare phrase structure. In Héctor Campos & Paula Kempchinsky (eds.), *Evolution and revolution in linguistic theory*, 51–109. Washington, D.C.: Georgetown University Press.
- Chomsky, Noam. 2004. Beyond explanatory adequacy. In Adriana Belletti (ed.), *Structures and beyond: The cartography of syntactic structures, volume 3*, 104–131. Oxford: Oxford University Press.
- Chomsky, Noam. 2008. On phases. In Robert Freidin (ed.), *Foundational issues in linguistic theory: Essays in honor of Jean-Roger Vergnaud*, 133–166. Cambridge, MA: MIT Press.

- Chung, Inkie. 2009. Suppletive verbal morphology in Korean and the mechanism of vocabulary insertion. *Journal of Linguistics* 45(3). 533–567.
- Collins, Chris & Erich Groat. forthcoming. Distinguishing copies and repetitions. In Kleantes K. Grohmann & Evelina Leivada (eds.), *The Cambridge handbook of the minimalist program*, Cambridge University Press.
- Collins, Chris & Richard S. Kayne. 2023. Towards a theory of morphology as syntax. *Studies in Chinese Linguistics* 44(1). 1–32.
- Corver, Norbert & Jairo Nunes (eds.). 2007. *The copy theory of movement*. John Benjamins.
- Embick, David. 2010. *Localism versus globalism in morphology and phonology*. Cambridge, MA: MIT Press.
- Fanselow, Gisbert & Damir Ćavar. 2001. Remarks on the economy of pronunciation. In Gereon Müller & Wolfgang Sternefeld (eds.), *Competition in syntax*, 107–150. Berlin: De Gruyter Mouton.
- Halle, Morris. 1997. Distributed Morphology: Impoverishment and Fission. In Benjamin Bruening, Yoonjung Kang & Martha McGinnis (eds.), *PF: Papers at the interface*, 425–449. Cambridge, MA: MITWPL.
- Halle, Morris & Alec Marantz. 1993. Distributed Morphology and the pieces of inflection. In Kenneth Hale & Samuel Jay Keyser (eds.), *The view from building 20: Essays in honor of Sylvain Bromberger*, 111–176. Cambridge, MA: MIT Press.
- Harris, Alice C. 2016. *Multiple exponence*. New York: Oxford University Press.
- Johnson, Kyle. 2012. Towards deriving differences in how *Wh* movement and QR are pronounced. *Lingua* 122. 529–553.

- Johnson, Kyle. 2020. Rethinking linearization. In András Bárány, Theresa Biberauer, Jamie Douglas & Sten Vikner (eds.), *Syntactic architecture and its consequences II: Between syntax and morphology*, 113–135. Berlin: Language Science Press.
- Johnson, Kyle. 2023. Keeping sisters straight. June 25. Paper presented at How Many Mothers? Multidominance in Syntax. Amherst: University of Massachusetts.
- Kandybowicz, Jason. 2004. Nupe tonology and the categorical identity of verb copy tones: A pilot experimental study. *University College London Working Papers in Linguistics* 16. 17–53.
- Kandybowicz, Jason. 2006. On Fusion and the architecture of PF. Paper presented at Penn Linguistics Colloquium (PLC) 30. Philadelphia: University of Pennsylvania.
- Kandybowicz, Jason. 2007. On fusion and multiple copy spell-out: The case of verbal repetition. In Norbert Corver & Jairo Nunes (eds.), *The copy theory of movement* (Linguistik Aktuell/Linguistics Today 107), 119–150. John Benjamins.
- Kandybowicz, Jason. 2008. *The grammar of repetition: Nupe grammar at the syntax-phonology interface*. Amsterdam: John Benjamins.
- Kayne, Richard S. 1994. *The antisymmetry of syntax*. Cambridge, MA: MIT Press.
- Kremers, Joost. 2012. The syntax of simultaneity. *Lingua* 122. 979–1003.
- Kremers, Joost. 2013. Linearisation as repair. In Patrick Brandt & Eric Fuß (eds.), *Repairs: The added value of being wrong*, 207–236. Boston: De Gruyter.
- Kusmer, Leland Paul. 2020. Optimal Linearization: Word-order typology with violable constraints. *Syntax* 23(4). 313–346.

- Lionnet, Florian, Laura McPherson & Nicholas Rolle. 2022. Theoretical approaches to grammatical tone. *Phonology* 39(3). 385–398.
- Malanoski, Aidan. 2024. Right node raising and Flexible Cyclic Linearization. In Jiaji Yu, Erika Petersen, Anissa Zaitso & Boris Harizanov (eds.), *Proceedings of the 40th West Coast Conference on Formal Linguistics*, 223–232. Somerville, MA: Cascadilla Proceedings Project.
- McDaniel, Dana. 1986. *Conditions on wh-chains*. New York: City University of New York PhD thesis.
- Nunes, Jairo. 2004. *Linearization of chains and sideward movement*. Cambridge, MA: MIT Press.
- Oda, Kenji. 2012. *Issues in the left periphery of Modern Irish*. Toronto: University of Toronto PhD thesis.
- Ostrove, Jason. 2018. Stretching, spanning, and linear adjacency in Vocabulary Insertion. *Natural Language & Linguistic Theory* 36. 1263–1289.
- Pfau, Roland & Josep Quer. 2002. V-to-Neg raising and negative concord in three sign languages. *Rivista di Grammatica Generativa* 27. 73–86.
- Prince, Alan & Paul Smolensky. 2004. *Optimality Theory: Constraint interaction in generative grammar*. Malden: Blackwell.
- Sayers, Peig. 2009. *Labharfad le cách*. New Island Publishing in cooperation with Raidió agus Teilifís na Éireann.
- Selkirk, Elisabeth. 2011. The syntax-phonology interface. In John Goldsmith, Jason Riggle & Alan C. L. Yu (eds.), *The handbook of phonological theory*, 435–484. Malden: Wiley-Blackwell 2nd edn.

- Sheehan, Michelle L. 2013. Some implications of a copy theory of labeling. *Syntax* 16(4). 362–396.
- Starke, Michal. 2009. Nanosyntax: A short primer to a new approach to language. *Nordlyd* 36(1). 1–6.
- Svenonius, Peter. 2016. Spans and words. In Daniel Siddiqi & Heidi Harley (eds.), *Morphological metatheory*, 201–222. Amsterdam: John Benjamins.
- Taraldsen, Knut Tarald. 2018. Spanning versus constituent lexicalization: The case of portmanteau prefixes. In Lena Baunaz, Liliane Haegeman, Karen De Clercq & Eric Lander (eds.), *Exploring nanosyntax*, 88–107. Oxford: Oxford University Press.
- Trommer, Jochen. 1999. Morphology consuming syntax’ resources: Generation and parsing in a minimalist version of Distributed Morphology. In Christian Retoré & Edward Stabler (eds.), *Proceedings of the ESSLI workshop on resource logics and minimalist grammars*, Utrecht: Association for Logic, Language, and Information.
- Vanden Wyngaerd, Guido, Karen De Clercq & Pavel Caha. 2021. Late Insertion and root suppletion. *ReVEL* 19(18). 81–123.
- de Vries, Mark. 2009. On multidominance and linearization. *Biolinguistics* 3(4). 344–403.
- Williams, Edwin. 2003. *Representation Theory*. Cambridge, MA: MIT Press.
- Woolford, Ellen. 2016. Two types of portmanteau agreement: Syntactic and morphological. In Géraldine Legendre, Michael T. Putnam, Henriëtte de Swart & Erin Zaroukian (eds.), *Optimality-Theoretic syntax, semantics, and pragmatics*, 111–135. Oxford: Oxford University Press.

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Notes

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¹Many linguists have used the term *antisymmetry* to refer to the property known in other fields as *asymmetry*. In this article, we hold to the standard usage of these terms, rather than the usage that has developed in linguistics. These properties are defined in section 2.

²A reviewer notes that in Nanosyntax, there is a sense in which this is false. For example, the same syntactic unit can receive different exponents depending on whether it is spelled out on its own or as part of a larger morpheme, pursuant to the Superset Principle. Thus, the mapping is not one-to-one *when comparing different derivations*. When considering the externalization of a single derivation, however, a given phrase is mapped onto a single lexical item, and lexical items do not overlap in which parts of the syntactic structure they realize. This is the relevant sense in which Nanosyntax assumes a one-to-one mapping.

³Nunes (2004) proposes an additional mechanism for multiple exponence. He assumes that re-Merge leaves copies, and that copies must be deleted in order for a syntactic structure to be linearizable. He further assumes that Fusion renders the fused nodes invisible to linearization. Because linearization is what motivates copy deletion, and Fusion makes nodes invisible to linearization, a moved constituent can be realized more than once if one of its copies undergoes Fusion: once on its own and then again as part of a fused morpheme.

⁴A reviewer notes that it is hypothetically possible that linearization generates a linearly ordered u-precedence re-

lation, with the linear ordering of the s-precedence relation parasitic on this. However, this approach is not clearly compatible with the present proposal. Suppose we have a u-precedence relation containing a single ordered pair $\langle X, X \rangle$, where X is a terminal node. This is trivially a linear order. Now suppose that Vocabulary Insertion inserts two exponents x_1, x_2 for X (see section 3.2). Then the s-precedence relation is $\{\langle x_1, x_1 \rangle, \langle x_1, x_2 \rangle, \langle x_2, x_1 \rangle, \langle x_2, x_2 \rangle\}$, which is not antisymmetric. In general, because the mapping from the u-precedence relation to the s-precedence relation, is not one-to-one, a linearly ordered u-precedence may be mapped onto an s-precedence relation that is not a linear order.

⁵Note that we can delete $\langle x, y \rangle$ or $\langle y, x \rangle$, but not both: if we delete both ordered pairs, then x will not be ordered with respect to y , so the s-precedence relation will not be total.

⁶Just as we have a Principle of Maximality, we could propose a Principle of Dependence that forbids the addition of ordered pairs to the s-precedence relation that do not correspond to ordered pairs in the u-precedence relation. However, the current framework provides no mechanism for such additions in the first place, so such a constraint is unnecessary.

⁷Our use of *improper* here invokes set theoretic terminology, where the term *improper subset* is sometimes used to characterize a subset that is equal to (rather than a proper subset of) its superset.

⁸In general, the converse would only hold if for every Vocabulary Item x inserted, there exists $\langle x, x \rangle \in P_s$. If precedence is read from the syntactic structure, this means that every terminal node would have to be in a reflexive relationship with itself.

⁹A reviewer asks how exponents of the same syntactic terminal(s) are distinguished, particularly if the exponents are phonologically identical. While we cannot offer a proposal on how to distinguish copies and repetitions within the space of this article (see Collins & Groat, forthcoming for issues surrounding the distinction between copies and repetitions within the Minimalist Program), we note that the same mechanism that distinguishes identical exponents that realize different syntactic terminals (e.g., the two occurrence of *who* in *Who sees who?*) will also suffice to distinguish identical exponents that realize the same syntactic terminal. After Vocabulary Insertion, the issue becomes how to distinguish (phonologically) identical exponents, not how to distinguish (featurally) identical syntactic objects—indeed, no syntactic information is encoded in the s-precedence relation—so there is no difference between “copies” and “repetitions” at this stage in externalization.

¹⁰In general, these factors will also favor smaller instances of multiple exponence (e.g., of heads rather than complex phrases).

¹¹In reality, the different surface forms may reflect different syntactic structures, rather than differences at PF. However, this is orthogonal to the present goal, which is to show how single and multiple exponence arise under this proposal.

¹²A reviewer asks how to determine where exponents surface. To some extent, this must be language-specific; see, for example, the different patterns of exponence discussed in Nunes (2004) and Corver & Nunes (2007). However, the distribution of exponents is constrained by the requirement that the s-precedence relation be a linear order, which effectively limits exponents to the possible surface positions of the occurrences of the constituents that they realize. In Figure 1, for example, realizations could hypothetically appear sentence-initially (realizing the occurrence in Spec,CP₁), between *du* and *Hans* (realizing the occurrence in Spec,CP₂) or sentence-finally (realizing the occurrence in Comp,V₂), but not elsewhere.

¹³We thank an anonymous reviewer for raising this point.

¹⁴We assume here that (i) Kandybowicz is correct that the tonal morpheme's need for a host triggers the multiple exponence of the verb and (ii) that PF processes such as Vocabulary Insertion cannot look ahead to the phonology, and thus cannot "know" that /L/ 'FOC' needs a host.

¹⁵The null morphemes *AgrO* and *v* are involved in various violations of antisymmetry (e.g., there is both $\langle \text{ba}_1, \text{AgrO} \rangle$ and $\langle \text{AgrO}, \text{ba}_1 \rangle$) which can be resolved in multiple ways (e.g., by deleting $\langle \text{ba}_1, \text{AgrO} \rangle$ or $\langle \text{AgrO}, \text{ba}_1 \rangle$ in the previous case). Because they are null and do not impact the string, it does not matter which of these possibilities is chosen, as long as the resulting s-precedence relation is a linear order. In practice, a null morpheme can be realized in any position in which the head it realizes occurs.

¹⁶Just as our account of multiple exponence raises the issue of how to distinguish different exponents of the same node(s) (see footnote 9), this account of portmanteaux raises the issue of how PF "knows" to map different terminals onto the same exponent, rather than to separate but identical exponents. Although this issue is beyond the scope of the present article, we note that for single exponence, this problem is trivial. To take the case of (23), if Vocabulary Insertion inserts only one instance of *bhfac*-, but Mood, $\sqrt{\text{SEE}}$ and *v* are all mapped onto *bhfac*-, then they must share the exponent.

¹⁷Embick (2010) argues that phonologically null material does not intervene in linear adjacency. None of the examples considered in this article weigh on this issue. However, if Embick is correct, this can be accommodated with minimal modification of our description of how the s-precedence relation is generated: for all $\langle X, Y \rangle \in P_u$, Vocabulary Insertion adds $\langle x, y \rangle$ to P_s for all *non-null* *x* inserted for *X* and for all *non-null* *y* inserted for *Y*.

¹⁸For the purposes of demonstrating crossover-based portmanteau, it is not crucial whether Neg is higher than Hon (Choi & Harley, 2019) or vice versa (Chung, 2009). However, if Hon is lower, then we can account for the possibility of honorific suppletion and the impossibility of negative suppletion in long-form negation (see footnote 20).

¹⁹Although Chung (2009) takes Neg to be lower than Hon, he also faces issues. For example, *iss*- 'exist' has both a negative suppletive form and an honorific suppletive form. If Neg is closer to the root than Hon is, then an approach

to suppletion based on structural adjacency would incorrectly predict the negative suppletive form *eps-* to surface in negative honorific contexts, when in fact the honorific suppletive form *kyeysi-* surfaces (Chung, 2009; Choi & Harley, 2019).

²⁰We have assumed here that the node realized as *-ci* heads some projection XP, and that head movement of Hon to Neg skips this head. This is not crucial. What is crucial is that the node realized as *-ci* (i) occupies a position where it will intervene linearly between Hon and Neg, and (ii) the node does not move past the base-merge position of Neg. This is necessary to account for the fact that negative suppletion is incompatible with long-form negation, but honorific suppletion is not (see section 3.5). In the specifier-initial, head-final linearization we have assumed here for expositional purposes, the only positions that satisfy these criteria are a head position or a right adjunct. Given a different linearization algorithm, other positions (e.g., a specifier position) may be available.

²¹Had $\sqrt{\text{KNOW}}$ moved to C, insertion of *molu-* would have been possible, although it would have been predicted incorrectly to follow *-ci* because $\langle X, \text{Neg} \rangle$ would still be protected by the Principle of Maximality.

²²A reviewer asks whether portmanteaux must be inserted when available. This question is beyond the scope of the present article: we are not proposing here a specific algorithm for Vocabulary Insertion, but rather arguing that it would be beneficial for such an algorithm to allow many-to-many mappings. However, we note that in certain cases, portmanteaux appear to be optional (Ostrove, 2018, fn. 6), so unless it can be shown that such apparent optionality reflects a difference in syntactic structure, it is desirable that Vocabulary Insertion allow for optionality in portmanteau. At the same time, other factors may favor the insertion of portmanteaux where possible. Inserting portmanteaux has the general effect of shrinking the s-precedence relation, since different ordered pairs in the u-precedence relation are mapped onto the same pairs in the s-precedence relation. This typically has the effect of reducing the number of ordered pairs that must be deleted to repair the s-precedence relation, so to the extent that the repair approach is correct and subject to economy, this may favor the insertion of portmanteaux. Additionally, Kremers (2013) suggests that in general, simultaneity is the default in linearization, with sequential representations only produced when simultaneous representations are unavailable. If this is correct, then we would expect portmanteaux to be inserted where possible.

²³Our proposal does not rule out that a contextual allomorphy approach to portmanteaux and suppletion may be appropriate in certain cases. In such an approach, an apparent portmanteau realizing some nodes *X* and *Y* is actually a contextual allomorph of *X* inserted in the context of *Y*, which may be null in the context of *X* (Trommer, 1999).

²⁴A reviewer notes that this is similar in spirit to superset-based spell-out. However, in our proposal, a portmanteau can be inserted for nodes that do not constitute a constituent or span, as demonstrated in the discussion of (24) above.

²⁵Vocabulary Insertion and linearization are independently motivated both empirically and conceptually. Empirically, they are independently motivated in that neither operation is proposed solely to account for portmanteaux. Con-

ceptually, they are independently motivated in that some mechanism is independently required to pair syntactic structure with phonological content (the role of Vocabulary Insertion), and to give this content temporal order (the role of linearization). While these mechanisms are independently motivated, it is not conceptually necessary that they be post-syntactic (e.g., Collins & Kayne, 2023)).

²⁶As a minor point, note that the negation marker *an(i)=* can be linearized in its surface position without a morphological operation of Lowering (Choi & Harley, 2019).

²⁷We thank Daniar Kasenov and an anonymous reviewer for raising these issues.

²⁸Distributed Morphology's Subset Principle (Halle, 1997) and Nanosyntax's Superset Principle (Starke, 2009) can be considered inviolable faithfulness constraints governing permissible mismatches between features of syntactic structures and features of their exponents. In an Optimality Theoretic approach, these would be replaced by one or more faithfulness constraints. For example, a form of MAX might penalize cases where an exponent has less features than the structure it realizes, while a form of DEP might penalize cases where an exponent has more features than the structure it realizes.

²⁹We intend an inclusive *or* here. For example, as understood here, Vocabulary Insertion takes as input both the output of syntax and the output of linearization.

³⁰This point holds regardless of whether Merge produces sets or ordered pairs. If Merge produces sets, then ordered pairs of terminal nodes have no status in the syntax, so the u-precedence relation cannot be a syntactic object. If Merge produces ordered pairs, then a complex syntactic object is an *ordered pair* of ordered pairs. The u-precedence relation, by contrast, is a *set* of ordered pairs, and thus not a syntactic object.